ROYAL CANADIAN NAVY
QUALIFICATION STANDARD AND PLAN (QSP)

Naval Warfare Officer (NWO) (00207)

Submarine Operation Room Officer (SMORO)

NQUAL - AILU
MITE Course ID – 122901

RECORD OF AMENDMENTS

The training authority (TA) designated for this qualification is the Commander Naval Personnel and Training Group (CNPTG). Naval Training Development Centre Atlantic (NTDC (A)) is the designated development centre responsible for the maintenance of training. Amendments to PO standard are to be requested through NTDC (A) and approved by CNPTG. Amendments to Lesson Guidance are to be approved by CO NTDC (A).

Revision	Approval Date	Authority
1.0	21 Feb 2022	CNPTG

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FOREWORD

1. This training is based on the specification found at :

   a. A-PD-055-002/PP-001

   b. A-PD-055-003/PQ-001

2. A trainee will achieve the qualification AILU when the standards listed in all Performance Objectives have been met.

3. A trainee will achieve the MITE Course ID 122901 when all Performance Objectives have been met.

4. The designated TEs for this training is indicated below:

TE	Phase / Module	MITE course ID	NQual
NFS(A)	All	122901	AILU

CHAPTER 1
TRAINING MANAGEMENT DETAILS

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AIM

1. The aim of this Qualification Standard and Plan (QSP) is to establish the requirements and training for the Naval Warfare Officer, Submarine ORO, AILU NQUAL. This QSP is designed to assist NTDC (A) in developing, Naval Fleet Schools in conducting and evaluating, and NPTG in validating the training.

COURSE DURATION

2. Course duration is calculated in accordance with (IAW) NPTORD 4500-22, Annex B. For this course, a period of instruction is 50 minutes in duration.

Course Duration Calculation	
Allocation of ATT	**Periods**
Sum of all EOs/POs	235
Enabling Checks (ECs)	88
Performance Checks (PCs)	122
Actual Training Time (ATT) Sub-total	445 55.7 **trg days**
Allocation of BAT, TAT, TTT and PT IAW NPTORD 4500-22, Table 5.5	
Base Administrative Time (BAT)	14
Training Administrative Time (TAT)	14
Training Travel Time (TTT)	nil
Physical Fitness (PT)	36
Sub-total	64 8 **trg days**
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **400 minutes** **Partial days must be rounded up**	64 **trg days**

REQUIRED COURSE STAFFING

3. Identify key personnel required to conduct the training.

Position	Rank	Qualifications	Qty	PO / EO
CI	LCdr	SMORO (AILU) Post Executive Officer/ (Preferred AEDM)	1	All
Junior Inst	Lt(N)/LCdr	AILU	1	All

COURSE CAPACITY

4. Planning for the conduct of training is based on the guidelines in CFITES Vol 5, Annex D, AP9-050-000/PT-005 and historical course numbers.

a. Maximum – 4;

b. Minimum – 2; and

c. Optimum – 3.

PREREQUISITES

Prerequisites. To be selected for this course, the loading authority must ensure that trainees have the following prerequisites:

a. Submarine Warfare Director (AILT), or Minor Warship Navigation (AEFF) qualified;

b. Recommended by Commanding Officer; and

c. Selected by Naval Board

COURSE REPORTING

Upon completion of the course, students shall receive a paper record of completion by means of form CF 377 Course Report and Form CF 289 (or locally produced equivalent) Certificate of Military Achievement IAW DAOD 31-9 and NPTORD 4500-51.

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

PLARs shall be conducted IAW NPTORD 4500-27, DAOD 5031-1 and CFITES Vol. 12.

SPECIAL INSTRUCTIONS

Nil.

CHAPTER 2

PERFORMANCE OBJECTIVES & LESSON GUIDANCE

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PO 001 – Plan an Alongside Submarine Program

1. **Performance**: Plan an Alongside Submarine Program

2. **Conditions**:

   a. Given:

      1) Master Implementation Plan Template;
      2) Calendar Template;
      3) Printer; and
      4) References.

   b. Denied: Nil

   c. Environment: At sea and alongside

3. **Standard**: The SMORO Student will plan a Alongside Submarine program (T 0028 and BT 0010) by:

   a. Extracting information from an operational schedule:

   b. Liaising with internal team to include all HODS and CHODS:

   c. Attending stakeholder meetings:

   d. Building a program to include (T0003 and T0007):

      1) Alongside calendar;
      2) Training program for crew;
      3) Tiered Readiness Programme;
      4) Licensing;
      5) Work-ups Programme; and
      6) Master Implementation plan.

   e. Brief Command (CT 0120)

4. **Training Limitations**: Nil.

EO 001.01 - EO Gather Information to Create Submarine Program

001.01A – Gather Info and OPSCHED

Conditions		Sequence
Classroom		
Time	**Classification**	**Instructor**
50 min	Confidential	Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	Method
1) Where to get information 2) How to access the CSF OPS SharePoint site on CSNI 3) Extracting information from an operational schedule	Think pair share

Scenario	Nil
Activity Steps	1) Students should have paper and pens. Students asked to think about what where they would gather information to produce a submarine program. 2) Each student will write down their ideas. 3) Get students to share ideas with a partner. Then discuss as a class. 4) Instructor will give correct information, and discuss answers. 5) Instructor to present templates of the MIP and Calendar and explain to students what the standard should be for the PC

Resources	References
1)Note books	1) CFCD 110 - SUBOPINST

2)Writing instruments 3) Master Implementation Plan Template 4) Calendar Template 5) UNCLASS DWAN Computer with Microsoft Office (one per student) 6) Printer	2) CFCD 129- RCN Readiness and Sustainment Policy 3) Submarine Workup Guide 4) NAVORD 1150-0 SUBSAFE 5) OPSCHED from CSF OPS 6) Applicable CFTOs (Trial program)

Notes to Instructor

- This lesson is meant to start off the PO 001 and provide an introduction for students in Planning an Alongside Submarine Program. Students should also be given their scenarios and resources for PC 001 in order to prepare
- The instructor should discuss the role of CSF OPS in the planning process and the division of responsibility between internal and external schedules

EO 001.01 - EO Collect Information to Create Submarine Program

001.01B – FTA/ST(S)/Technical Officer Lecture

Conditions		**Sequence**
Classroom/Campus		After EO 001.01A
Time	**Classification**	**Instructor**
50 min	UNCLAS	NTO: - FTA-SUBS SSO/ ST(S)/ Submarine HOD MSEO and CSEO currently employed in the role or post HOD NWO: - Min Qualification: AILU - Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	**Method**
1) Understand the relationship between Sub Staff and outside organizations with regards to building an Alongside Program 2) Understand the key dates and requirements with regards to Submarine Licensing 3) Understand the Submarine Licensing process to include Tiered Readiness Requirements from level 1 to 3 4) Understand the required ST(S) serials and the reason for each and the steps required and how they fall into licensing	Guided Discussion

Scenario	Nil
Activity Steps	The guest instructor is to discuss with the students the following: - The roles of FTA/FMF/NETE/and other technical organizations in how they affect the OPSCHED - Submarine Licensing and what the levels mean and specially how to achieve each level - How and why are technical delays and how they affect they OPSHED (such as a delayed undocking date)

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	- Using the CLASS plan from DMEPM-SM and discuss what the different work periods and what the scope of work for each on it. - Using the SWG explain the terms for a Submarine Workup and how they tie into licensing - List some required inspections and DAG procedure that is required in an Alongside Program

Resources	References
1)Note books 2)Writing instruments 3) Master Implementation Plan Template 4) Calendar Template 5) UNCLASS DWAN Computer with Microsoft Office (one per student) 6) Printer 7) Guest Speaker as per instructor field	1) CFCD 110 - SUBOPINST 2) CFCD 129- RCN Readiness and Sustainment Policy 3) Submarine Workup Guide 4) NAVORD 1150-0 SUBSAFE 5) OPSCHED from CSF OPS 6) Applicable CFTOs (Trial program)

Notes to Instructor

- There will be no test of knowledge at the end of this EO. The PC project is the assessment criteria. Guest instructor is encouraged to use PowerPoint and to also use the most recent CLASS PLAN from DMEPM-SM and explain to students in detail the material as this will be a brand new subject for the students.

EO 001.01 - EO Collect Information to Create Submarine Program

001.01C – CFMWC OPP Lecture

Conditions		Sequence
CFMWC		After EO 001.01A
Time	**Classification**	**Instructor**
800 min	Unclassified	Guest Lecturer: CFMWC for OPP only.

Learning Objectives	Method
1)OPP	Lesson and syndicate work

Scenario	Nil
Activity Steps	As per CFMWC curriculum.

Resources	References
1)Book guest	1) IAW CFMWC curriculum

Notes to Instructor
Nil

EO 001.02 – Build an Alongside Programme

001.02A – Create an Alongside Calendar

Conditions		Sequence
Classroom/Campus		After EO 001.01
Time	**Classification**	**Instructor**
100 min	UNCLAS	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	Method
1) How to fill out an alongside calendar	Case Study Guided Discussion

Scenario	Students will be given a scenario that is shorter in duration than their PC. They are to complete an Alongside Calendar and submit to the instructor for feedback as an EC. Assessment criteria is one that is used as per PC 001
Activity Steps	- The instructor will give each student a scenario and have them complete an Alongside Calendar - The instructor will provide the template and the standard that will be considered satisfactory - The instructor will also show students tips on how to efficiently populate an Alongside Calendar - The length and duration of the assignment should not be as long as the PC. The goal is to ensure that the student can start and create a Calendar. Not if they can create a full calendar from undocking to deployment. The EC should be ideally concentrate on only a few legs of alongside time - Students will get the remainder of 100mins classroom time to produce EC and may also work after hours

Resources	References
1)Note books 2)Writing instruments 3) Master Implementation Plan Template 4) Calendar Template 5) UNCLASS DWAN Computer with Microsoft Office (one per student) 6) Printer	1) CFCD 110 - SUBOPINST 2) CFCD 129- RCN Readiness and Sustainment Policy 3) Submarine Workup Guide 4) NAVORD 1150-0 SUBSAFE 5) OPSCHED from CSF OPS 6) Applicable CFTOs (Trial program)

Notes to Instructor

- Assign a case study to each student and have them complete at most 3 legs of each alongside period. The goal is to see if the student and correctly plan multiple events and de-conflict appropriately. If they have trouble they are not denied help

EO 001.02 – Build an Alongside Program

001.02B – Create a Master Implementation Plan

Conditions		Sequence
Classroom/Campus		After EO 001.02A
Time	**Classification**	**Instructor**
50 min	UNCLAS	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	Method
1) How to create a Master Implementation Plan	Case Study Guided Discussion

Scenario	Students will be given a scenario that is shorter in duration than their PC. They are to complete an MIP for one event (such as undocking, sailing, HLV) and submit to the instructor for feedback as an EC. Assessment criteria is one that is used as per PC 001
Activity Steps	- The instructor will give each student a scenario and have them complete an MIP - The instructor will provide the template and the standard that will be considered satisfactory - The instructor will also show students tips on how to efficiently populate an MIP including all required tasks - The length and duration of the assignment should not be as long as the PC. The EC shall be only for only one major task - Students will get the remainder of 50 mins classroom time to produce EC and may also work after hours

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Resources	References
1)Note books 2)Writing instruments 3) Master Implementation Plan Template 4) Calendar Template 5) UNCLAS DWAN Computer with Microsoft Office (one per student) 6) Printer	1) CFCD 110 - SUBOPINST 2) CFCD 129- RCN Readiness and Sustainment Policy 3) Submarine Workup Guide 4) NAVORD 1150-0 SUBSAFE 5) OPSCHED from CSF OPS 6) Applicable CFTOs (Trial program)

Notes to Instructor

- Assign a case study to each student and have them complete 1 MIP prior to a significant event (Undocking, HLV, Camber Dive). The goal is to see if the student and correctly populate the MIP, come up with required tasks, and properly liaise with outstations to find out what tasks should be added to the MIP. If they have trouble they are not denied help

EO 001.03 – Brief Command on Alongside Calendar

001.03A – Brief Command

Conditions		Sequence
Classroom/Campus		After EO 001.02B
Time	**Classification**	**Instructor**
100 min	UNCLAS	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	Method
1) How to create a Alongside Program Presentation and Brief Command	Guided Discussion Fishbowl

Scenario	Using the previously assigned EC 001.02A and EC 001.02B students are to create a Power Point presentation to Brief Command on their Alongside based on the assessment criteria that is used as per PC 001
Activity Steps	- The instructor gives each student a template of a Command Brief on an Alongside Program and demonstrates to the students what is the standard that is expected - This is broken up into two 50 min periods. One period is a guided discussion and creation of brief. The next period is used for presenting the Brief to the instructor

Resources	References
1)Note books 2)Writing instruments	1) CFCD 110 - SUBOPINST

3) Master Implementation Plan Template 4) Calendar Template 5) UNCLASS DWAN Computer with Microsoft Office (one per student) 6) Printer	2) CFCD 129- RCN Readiness and Sustainment Policy 3) Submarine Workup Guided 4) NAVORD 1150-0 SUBSAFE 5) OPSCHED from CSF OPS 6) Applicable CFTOs (Trial program)

Notes to Instructor

- Assign having gone over each students EC Calendar and MIP have them compile a Command Brief. Each student will brief their presentations with the other students observing for learning purposes

PC 001- Plan an Alongside Submarine Program

Scope	Conditions	Sequence
PO 001	Classroom	- Conducted after OPP - Conducted after guest lectures from (FTA, ST(S))
Time	**Classification**	**Instructor**
- 600 mins	UNCLAS	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Scenario	The student is to act as the submarine's Operations Officer and must produce an Alongside Program given information by the instructor. The submarine is currently finishing a docking work period and needs to conduct a Tiered Readiness Program to bring itself from Extended Readiness up to High Readiness 1 in order to be prepared for an upcoming deployment. The student will be expected to produce a Calendar and a Master Implementation Plan that will highlight all key events and requirements to get to High Readiness 1. The student will present this to the instructor as a hand in as well as prepare a Command Brief. Student will be given the PC scenario at the beginning of the PO. Can work on PC as lessons progress. After last lesson student given one day (400min class time) to complete PC. Next morning students will give command brief (One period per student; total 200 min) The instructor will act as the submarine Commanding Officer and applicable outstations. The student is allowed to seek assistance from third parties such as FTA, ST(S), and CSF OPS.
Assessment Steps	1) The Student is given: a. OPSCHED b. References listed below c. Contact information for third parties

	d. Scenario – PC scenario given at beginning of PO. 2) The Student is expected to produce: a. An alongside Calendar outlining key dates b. A Master Implementation Plan with required tasks to affect events c. A brief to Command highlighting key events and dates 3) The Instructor is expected to provide: a. A draft OPSCHED b. Act as third parties to the students for training value c. All references listed below d. Computer and printing capabilities with Microsoft Office 4) The output of the work is: a. Assessed using a checklist as listed in the assessment criteria 5) The completed products are: a. Submitted to the instructor for filing

Resources	References
1) Master Implementation Plan Template 2) Calendar Template 3) UNCLASS DWAN Computer with Microsoft Office (one per student) 4) Printer	1) CFCD 110 - SUBOPINST 2) CFCD 129- RCN Readiness and Sustainment Policy 3) Submarine Workup Guide 4) NAVORD 1150-0 SUBSAFE 5) OPSCHED from CSF OPS 6) Applicable CFTOs (Trial program)

Notes to Instructor

- This assessment is given as a project and is usually given a week to complete
- The students should be given all applicable resources as possible including the ability to contact third parties for real world guidance
- The students will hand in their project and the instructor will review based on the assessment criteria and review with the students as required
- The students will also provide a brief to the instructor covering key topics and dates

Limitations	N/A The student is encouraged to seek out as many resources and ways to conduct this task as possible
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

Item	PC 001 Plan an Alongside Submarine Program	SAT	UNSAT	Remarks
1. Build an Alongside Calendar				
Key Dates/Commitments Identified and Extracted from OPSCHED as required: - Undocking - Camber Dive - National Exercise Commitments - Deployment Dates		☐	☐	
Internal Training Program Implemented as required: - OTT1/2, WST 1/2, MCT/SCT, IPTT, Bling Pilotage Training - Refresher Training (FF, CBRN, Small Arms - DAG		☐	☐	
Liaised with HODs/CHODs to include key events such as required: - Fueling - Ammunitioning - Inspections (Nav Admin, Health and Welfare, Logistics Compliance, Ammunitions, Escape)		☐	☐	
Included information from external stakeholders to include events: - SUBSAFE Licensing Requirements for Level 1,2,3 - Sea Training (Submarines) Harbour Liaison Visit, Harbour Readiness Training (HLV/Alongside Workups Program) - Tiered Readiness Program - Work Ups dates - Sea Trials - Weapons Certification		☐	☐	
Potential conflicts identified in scheduling and alternative plan recommended		☐	☐	

2. Build a Master Implementation Plan			
Key tasks listed to affect important dates	☐	☐	
Multiple tabs/sections divided by Department	☐	☐	
Tracking system in place to measure percentage complete	☐	☐	
3. Brief Command			
Key dates discussed to include: - What is required in the MIP to affect key dates	☐	☐	
Potential scheduling conflicts presented with recommended solutions	☐	☐	
Presentation clear and concise	☐	☐	

Score: /11

Trainee must achieve 11/11 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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PO 002 – Create a Duty Plan for a Sailing Period

1. **Performance**: Create a duty plan for a sailing period.

2. **Conditions**:

a. Given:

(1) Duty Plan template;
(2) Access to Trinity resources;
(3) CSNI access; and
(4) References

b. Denied:

(1) Nil

3. **Environmen**t:

c. At sea or alongside

4. **Standard**: The SMORO student shall create a submarine duty plan for a sailing period by (BT 0050,BT0030,T0029):

d. Extracting information from higher authority:

e. Creating threat package (T0019);

f. Completing a duty plan to include (T0013):

(1) Situation;
(2) Mission;
(3) Priorities;
(4) Timings;
(5) Environment & Ephemerals (T0022);
(6) Execution to include;
    (a) Primary plan; and
    (b) Alternate plan.
(7) Navigation
    (a) Fixing Routine;
    (b) Actions Reaching Limiting Danger line;
    (c) Actions Reaching Safe Water Limit;
    (d) Areas;
    (e) Ephemerals
    (f) Tidal data;
    (g) HOT;
    (h) Charts in use:
    (i) Other references;

(j) Depth Information;
(k) Alarm Sounding;
(l) Safe water limit;
(m) ES policy;
(n) MDD;
(o) Holding area;

(8) Modified Ship Control;
(9) IARL;
(10) Weapon Status;
(11) Discussions Points to include;
(a) Technical;
(b) Tactical;
(c) Contacts
(d) ESM;
(e) Sonar;
(f) Visibility;
(g) Navigation;
(h) Personnel;
(12) Command Appreciation;
(13) Specific Aims;
(14) Threat/Intelligence to include (T0019);
(a) Military Traffic;
(b) Expected Civilian/Neutral Traffic
(15) Rules of Engagement/ CASEX Instructions (T0020);
(16) Expected Threat Actions/Reactions to include;
(a) Their operations (ECOA);
(b) In event of detection
(17) Own Reactions;
(18) Sensor Employment;
(19) Stern Arc Clearance;
(20) Approach Limitations;
(21) Platform Statues to include;
(a) Weapon Settings;
(b) SSE state;
(c) BATCAP required;
(d) Snorting plan;
(e) Readiness state;
(f) Watch changeover; and
(g) Meals.
(h) Records;
(i) EOPS;
(j) Preparations/Training;
(k) Communications.

g. Present Mission Brief:
h. Creating an Operations Package to include;
(1) Flex; and
(2) Operational messages to include (BT 0130);

(a) CASEX TABORD
(b) RSS
(c) RFI (N2/TRINITY)
(d) OPSTAT UNIT
(e) ROE request
(f) Starred relaxations request

i. Preparing evaluation operational reports (BT0140)

5. **Training Limitations**: Nil

EO 002.01 – Produce Recce/Threat Package

002.01A – Applicable Threats

Conditions		**Sequence**
Classroom		N/A
Time	**Classification**	**Instructor**
50 Mins	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	**Method**
1) Based on an OPORD produce a list of applicable military threats involved in an operation	Class Room Table Top Assignment

Scenario	Students to be given a country currently considered as hostile
Activity Steps	1) Provide Students with an OPORD for a particular military operations which identifies a clear hostile force 2) Provide instruction on the best means to determine applicable threats 3) Provide discussion on how to determine priority contact for air, surface, or sub-surface contacts.

Resources	**References**
1) Trinity 2) CSNI database	1) Jane's Fighting Ships 2) Jane's Online Sentinel Security Library (accessed through the Canadian Armed Forces Virtual Library via DWAN only) https://www.cfc.forces.gc.ca/188/223-eng.html

Notes to Instructor
Nil

EO 002.01 – Produce Recce/Threat Package

002.01B – Threat Research

Conditions		**Sequence**
Classroom		N/A
Time	**Classification**	**Instructor**
50 Mins	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	**Method**
-Compile Applicable Military Data for Threat list determined in preceeding lesson.	Class Room Table Top Assignment

Scenario	Students to be given a country currently considered as hostile
Activity Steps	1) Disucss the role of SonarOPS and NCIOPS in the compellation of radar and sonar information 2) Discuss detection and counterdection ranges for all applicable sensors 3) Discuss the role of the OPSO in the discussion of threats. In particular how we expect the threat to prosecute a submarine threat and what actions they will take if the unit is able to detect a submarine threat. 4) Discuss considerations on how the OPSO will advise command on how to remain covert and how to take offensive action against a threat.

Resources	**References**
1) Trinity 2) CSNI database	1) Jane's Fighting Ships

	2) Jane's Online Sentinel Security Library (accessed through the Canadian Armed Forces Virtual Library via DWAN only) https://www.cfc.forces.gc.ca/188/223-eng.html

Notes to Instructor
Nil

EO 002.01 – Produce Recce/Threat Package

002.01C – Brief Command on an applicable Threat

Conditions		**Sequence**
Classroom		N/A
Time	**Classification**	**Instructor**
100 Mins	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	**Method**
1)Present a threat brief as you would prior to an operation	Class Room Table Top Assignment

Scenario	1)Students to be given a country currently considered as hostile
Activity Steps	1) Lead a discussion on how to properly present a particular threat to command. 2) Provide an example to the students on how to properly brief a threat to command. If available it is preferable to have a SONAROP and NCIOP available to discuss their portions of a threat brief and provide insight for any particular threats. 3)The last phase here will involve the students practicing the methods used to brief a threat. Students will be provided time to consider the threat and then present to the instructor in preparation for the final PC 4) The presentation will be the EC.

Resources	References
1) Trinity 2) CSNI database 3.) Sonar OP or NCIOP to assist	1) Jane's Fighting Ships 2) Jane's Online Sentinel Security Library (accessed through the Canadian Armed Forces Virtual Library via DWAN only) https://www.cfc.forces.gc.ca/188/223-eng.html

Notes to Instructor

EO 002.02 – Produce a Duty Plan

002.02A – Discuss the purpose behind a duty plan

Conditions		**Sequence**
Classroom		N/A
Time	**Classification**	**Instructor**
50 Mins	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	**Method**
1)Provide an overview of the importance of a duty plan	Class Room Table Top Assignment

Scenario	Students will be provided with a proposed operation
Activity Steps	1) Students will be provided with an overview of the importance of a duty plan and the expectation from command concerning the duty plan 2)It would be preferable to have a SMCC qualified Officer for this discussion

Resources	**References**
1) SMCC	1) Norwegian Duty Plan Format

Notes to Instructor

EO 002.02 – Produce a Duty Plan

002.02B – Discuss Duty Plan format and content

Conditions		**Sequence**
Classroom		N/A
Time	**Classification**	**Instructor**
150 Mins	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	**Method**
1)Provide an overview of the importance of a duty plan	Class Room Table Top Assignment

Scenario	Students will be provided with a proposed operation
Activity Steps	1.) Students will be provided with an approved copy of a duty plan to discuss the format and the overall content of a duty plan 2.) The instructor will discuss each of the points in detail and clarify Command expectations for each point 3.)It would be preferable to have a SMCC qualified Officer for this discussion

Resources	**References**
1) SMCC Qualified Officer	1) Norwegian Duty Plan Format

Notes to Instructor

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EO 002.02 – Produce a Duty Plan

002.02C – Create a Duty Plan

Conditions		**Sequence**
Classroom		N/A
Time	**Classification**	**Instructor**
100 Mins	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	**Method**
1)Present a threat brief as you would prior to an operation	Class Room Table Top Assignment

Scenario	Students will be provided with a proposed operation
Activity Steps	1.) Students will be expected to produce a first draft of a duty plan 2.) The students will be given a scenario and asked to produce a duty plan for that operation. 3.)Classroom time to be devoted to its development as well as feedback sessions from the instructor to allow improvement and continued input throughout this process

Resources	**References**
1) Trinity 2) CSNI database 3.) Sonar OP or NCIOP to assist	1) Jane's Fighting Ships 2)

Notes to Instructor

EO 002.03 – Mission Brief

002.03A – Mission Brief Format

Conditions		**Sequence**
Classroom		N/A
Time	**Classification**	**Instructor**
50 Mins	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	**Method**
1)Present a threat brief as you would prior to an operation	Class Room Table Top Assignment

Scenario	Students will be provided with a proposed operation
Activity Steps	1.) Instructor will provide an overview of the content expected from command for a mission brief. 2.) This brief will rely heavy on the material created for the duty plan

Resources	**References**
1) Confidential laptop 2) CSNI	1) 2)

Notes to Instructor

EO 002.03 – Mission Brief

002.03B– Create a Mission Brief

Conditions		**Sequence**
Classroom		N/A
Time	**Classification**	**Instructor**
400 Mins	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	**Method**
1)Present a threat brief as you would prior to an operation	Class Room Table Top Assignment

Scenario	Students will be provided with a proposed operation
Activity Steps	1.) Students will create a mission brief and then schedule a feedback session with the instructor so input can be provided.

Resources	**References**
1) Trinity 2) CSNI database 3.) Sonar OP or NCIOP to assist	1) Jane's Fighting Ships 2)

Notes to Instructor

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PC 002 Create a Duty Plan for a Sailing Period

Scope	Conditions	Sequence
PC 002	Classroom	After Lesson 002.0X
Time	**Classification**	**Instructor**
-1000 mins to complete write-up	Confidential	Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Scenario	Submarine will be proceeding to sea for AMSRT. Student will be employed as an Operations Officer and create a duty plan for this sail. This plan will encompass all the planning and warfare aspects necessary for this period. This will include but not be limited to guidance for attack, weapons employment, counter detection, environmentals, detailed tasks, etc. Student will be given the PC scenario at the beginning of the PO (Students will select their threat, each student to have a different threat approved by instructor). Students will complete components of the PC as ECs. Revised ECs will be submitted for the PC. (800min class time) to complete PC. The next morning students will give PC command brief (One period per student; total 200 min) Students will present brief with fellow students present. Fellow students will not ask question or give comments.
Assessment Steps	1.) Provide student with a TASK ORDER, OP ORDER, OPSCHED, Daily Planner template, and enemy ORBAT 2.) Produce a threat pack, comprehensive duty plan, give command brief, operation package. 3.) Instructor will review duty plan and provide feedback which will be utilized to further develop existing plan 4.) Students will produce a final duty plan which will <u>be assessed</u> by the instructor, or if available, a SM Commanding Officer 5.) Create flex for sailing period and Daily planner 6.) Students will retain duty plan for use after the course

Resources	References

1) CSNI Access (each student should have their own CSNI account and access to a CSNI terminal) 2) Classified Stand Alone laptop with associated printer 3) USB Memory stick for transferring information between CSNI and laptops 4) Presentation hardware either a projector or applicable Large Screen 5) Access to CSNI Trinity Resources (each student should have their own CSNI account and access to a CSNI terminal)	1) SOCT Planning Guided 2) NCPM 411 STOPS 3) CFCD 106 4) CFCD 130 5) MC362 (NATO ROE Manual) 6.) CF USE OF FORCE MANUAL 7) MK48 USER MANUAL (FMS Vol 1 & 2) 8) Submarine Records Manual 9)CFCD 102 RCN CRR

Notes to Instructor
-Provide Duty Plan Templates and previous examples -Lectures detailing the necessary information

Limitations	N/A The student is encouraged to seek out as many resources and ways to conduct this task as possible
Impact of Failure	If the student does not satisfactorily complete the PC they will be given another scenario and will repeat it. Possible TRB in according to NPTORD.

Item	PC 002 Create a Duty Plan for a Sailing Period	SAT	UNSAT	Remarks
1. Create Recce Pack				
Identified Applicable Threats/Refs		☐	☐	
Identified ASW Sensors and Weapons		☐	☐	
Provided Ranging Heights		☐	☐	
Considered Threat ASW Actions		☐	☐	
Submarine Response Plan to Threat to include a. Evasion Plan b. Attack Plan c. Mission Priorities		☐	☐	
2. Create Duty Plan				
Provide Mission Statement		☐	☐	
Provide Limitations and Considerations		☐	☐	
Identify Priorities		☐	☐	
Identify significant Situational factors		☐	☐	
Provide Oceanography assessment		☐	☐	
Identify Navigational concerns		☐	☐	
Research Environmental Factors		☐	☐	
Research Civilian/Neutral Traffic		☐	☐	
Compile Friendly and Enemy Units		☐	☐	
Evaluate Enemy Forces and COAs		☐	☐	
ROE		☐	☐	
Execution plan to include; a. Use of Sensors b. Use of Masts c. Primary and Secondary Plan		☐	☐	
Snorting Plan		☐	☐	
Stop Snorting Criteria		☐	☐	
Calculate IARL and SAC		☐	☐	
Battery Considerations		☐	☐	
Communications Plan		☐	☐	
Weapons Plans		☐	☐	
Counter Detection Criteria		☐	☐	
Counter Detection Reactions		☐	☐	

Service and Support	☐	☐	
Command and Control	☐	☐	
Detailed Tasks	☐	☐	
3. Command Brief			
Create Sailing Period Flex	☐	☐	
Present Mission Brief to Command including the following:			
a. Mission Statement	☐	☐	
b. Submarine Priorities	☐	☐	
c. Area of Operation	☐	☐	
d. Situation Statement	☐	☐	
e. Applicable ROE	☐	☐	
f. Oceanographic Summary	☐	☐	
g. Use of Sensors and Masts	☐	☐	
h. Snort Plan	☐	☐	
i. Battery Considerations	☐	☐	
j. Weapon Considerations	☐	☐	
k. Counter Detection Criteria	☐	☐	
l. Service and Support	☐	☐	
m. Communications	☐	☐	
n. OPDEFS	☐	☐	
o. Safety Considerations	☐	☐	
Post Sail Reports: - Post Deployment Report - Quicklooks - First Impression Reports - OOW Narrative - General Operations Plot	☐	☐	
Score: / 45 The trainee must achieve 45 / 45 SATs to pass this assessment			
Positive Feedback	**Improvement Feedback**		

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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PO 003 – Run a Dive Watch in Support of Force Generation Operations

1. **Performance**: Run a dive watch in support of force generation operations.

2. **Conditions**:

   a. Given:

      a. A submarine at sea or the Fred Sherwood Command Team Trainer ;
      b. Command direction; and
      c. References.

   b. Denied: Nil

   c. Environment: At sea in any weather condition

3. **Standard**: The SMORO student shall Run a Dived Watch in support of Force Generation operations to include:

   a. Diving the submarine (T 0002):

   b. Maintaining Safety by:

      a. Interpreting the target Motion Analysis (TMA) Plots (T 0024)
      b. Directing EOPS (T 0005);
      c. Advising Command on UWW Safety (BT 0340);
      d. Advising Command on Anti-Surface Warfare Safety (BT 0330);
      e. Advising Command on Anti-Surface warfare safety (BT 0330);
      f. Executing periscope safety techniques eyes-only attack (T 0026);
      g. Assigning contact tracking priorities (T 0025)

   c. Conducting Force Generation operations to include;
      a. Direct tactical maneuvering (T 0009) to include;
         (a) Applying towed array tactics; and
         (b) Implement Anti-surface warfare tactics (BT 0300);
         (c) Implementing UWW Tactics (BT 0310)

      b. Managing UWW Operations (BT 0110);
      c. Interpreting meteorological oceanographic and acoustic environmental info (T 0022);
      d. Directing combat information organization (T 0017) include;
         (a) Directing combat system line-ups (BT 0110); and
         (b) Directing personal;
      e. Managing Anti-Surface warfare Operations (BT 0100);

   d. Completing post watch requirements to include;
      a. Exercise records (Quicklook Msg, FIR, OOW narrative); and
      b. General Operations Plot (GOP).

e. Surfacing the submarine (T 0001)

4. **Training Limitations**: Diving submarine training will include surfacing the submarine (T 0001)

EO 003.01 - Diving the Submarine

003.01A - Diving the Submarine

Conditions		**Sequence**
Classroom		
Time	**Classification**	**Instructor**
100mins	Unclassified	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	**Method**
1) Dive the Submarine as the role of the OpsO 2) Surface the Submarine in the role of the OpsO	Guided Discussion Table Topics

Scenario	Nil
Activity Steps	1) Students are given references below 2) Students are to be assigned either Diving or Surfacing the submarine and are to produce a Briefing Note on the process of how to dive or surface the submarine as the role of the Dived OOW/OpsO. They are to discuss the process for: a. The role of OpsO at Diving Station/Watch Diving on how to bring the submarine from the surfaced mode to dived mode including Open Up For Dive, Pre Diving Checklists, Diving (opening of Main Vents to change the buoyancy of the submarine), Post Diving checks, Fall out to Watch Dived b. The role of OpsO at Watch Dived/Standing by to Surface on how to bring the submarine from the dived mode to surfaced mode including Standing by to surface, Surfacing, Falling out to Surfaced Routines. Tactical Surfacing shall also be discussed in the briefing note to include operation considerations and NOGO criteria 3) Upon completion students will present their briefing notes to the instructor 4) Instructor will give correct information, and discuss answers. 5) Student should also include the following in their briefing notes: a. ARL interval and their calculation based on Visibility and known vessels in the area b. Considerations on tactical posture post diving checks/post surfacing c. Tactical Surfacing and what method to surface

Resources	**References**
1) UNCLASS DWAN Computer with Microsoft Office (one per student)	1) VCSOPs, VCEOPS, VCSSOs 2) CFCD 110B 3) NCPM 401 (B)

2) Printer for UNCLASS DWAN Computer 3) Classroom	4) NCPM 411 STOPs

Notes to Instructor

Completion of the Briefing Note will constitute the EC of this EO

EO 003.01 – Diving the Submarine

003.01B – Maintaining Safety by directing EOPS

Conditions		**Sequence**
Classroom		
Time	**Classification**	**Instructor**
50 mins	Unclassified	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	**Method**
1) Maintaining Safety by directing EOPS	Guided Discussion

Scenario	Nil
Activity Steps	1) Students are given references below 2) Students are to discuss the process for: a. The role of OOW (Dived) during EOPs. The students should be able to discuss with the instructor the following: i. Operational losses of the EOP ii. Adjustments that need to be made iii. Tactical limitations on the submarine iv. Maintaining the tactical picture/situational awareness while the SCOOW carries on with EOPs 3) The instructor will then give guidance and feedback

Resources	**References**
1) UNCLASS DWAN Computer with Microsoft Office (one per student) 2) Printer for UNCLASS DWAN Computer 3) Classroom	1) VCSOPs, VCEOPS, VCSSOs 2) CFCD 110B 3) NCPM 401 (B) 4) NCPM 411 STOPs

Notes to Instructor
This EO will not have an EC

003.02 – Return to Periscope Depth

Conditions		**Sequence**
Fred Sherwood Command Team Trainer		After EO 003.01
Time	**Classification**	**Instructor**
2000 mins (500 mins per student)	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	**Method**
1) Interpreting Target Motion Analysis 2) Assigning contact tracking priorities 3) Safely return the submarine to periscope depth from safe depth 4) Directing combat information organization to include; a. Directing combat system line-ups ; and b. Directing personnel	Practical

Scenario	Students will be in the Fred Sherwood Command Team Trainer in the role of OOW. They will be given resources and references as listed below. They will be given a scenario similar to an OTT1 run where they will need to return the submarine to periscope depth during a peacetime scenario with light to moderate contact density.
Activity Steps	1) Students are given references and resources below 2) Students will be given a run similar to OTT1: a. Peacetime in local waters (MARLOAS or SOG) b. Light to Moderate contact density c. No expected OpFOR 3) The students will then be expected to return the submarine to periscope depth 4) The instructor will then give guidance and feedback

Resources	**References**
1) Fred Sherwood Command Team Trainer 2) Control Room Team (Recommended 3x Sonar OPS, 2X Fire Control Operators, 1X 2OOW) 3) MLD	1) VCSOPs, VCEOPS, VCSSOs 2) CFCD 110B 3) NCPM 401 (B) 4) NCPM 411 STOPs 5) AXP 1 6) AXP 5 7) ATP 1 8) ATP 28

	9) CSOs

Notes to Instructor

Each student will receive at a minimum 5 runs (each run will consist of 2x50 min periods). The instructor will assist the students on a decreasing scale with heavy guidance during runs 1 & 2, light guidance during run 3 & 4, and almost no guidance during run 5

The 5th run will be the EC and the instructor will grade on a pass/fail on whether the student possess the basics of returning a submarine to periscope depth

This is a critical EC and the below checklist is to be used for assessing

Item	**EC 003.02 – Return to Periscope Depth** Run a Dived Watch in Support of Force Generation Operations	SAT	UNSAT	Remarks
1. Submarine Routine/State				
Conduct Plot Brief to include: - Situation/Geography - Mission - Team Input - Sensor Employment		☐	☐	
Conduct Watch/Outstation Brief		☐	☐	
2. Situational Appreciation				
Maintenance of Priorities		☐	☐	
Manoeuvre Submarine		☐	☐	
Assigning contact tracking priorities		☐	☐	
Conduct Drills and Reports IAW NCPM 401		☐	☐	
Demonstrate understanding of Submarine Manoeuvring Limitations: - MLD - Towed Array		☐	☐	
3. Tactical Appreciation				
Adopt IARL as appropriate		☐	☐	
Calculate LI		☐	☐	
React to environment as appropriate		☐	☐	
4. Safety				
Fishing Vessel Avoidance *** - Identify - Manoeuvre IAW CFCD 110		☐	☐	
Close Quarters Contact Situation *** - Identify - Manoeuvre IAW CSOs		☐	☐	
Navigation *** - POE IAW CFCD 130 - WSM/PMI		☐	☐	

Supervise SCOOW - Understand System Losses	☐	☐	
5. Periscope Skills			
Understand Situational Awareness*** - Contact Avoidance	☐	☐	
Conduct periscope math	☐	☐	
Use masts effectively to avoid counter detection (Mast Exposure)	☐	☐	
Conduct Standard Drills & Reports IAW CANSADS & NCPM 401***	☐	☐	
Understand Time Management	☐	☐	
6. TBP Analysis			
Contact Management	☐	☐	
Initial Target Solution	☐	☐	
Use of Formulae	☐	☐	
TBP Analysis	☐	☐	
Team Coordination	☐	☐	
Interpretation of B° Curves	☐	☐	
ZIG Recognition	☐	☐	

Score: / 26

The trainee must achieve 26 / 26 SATs to pass this assessment

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One

Met Standard	Did Not Meet Standard
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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EO 003.03 – Run an Eyes Only Attack

Conditions		Sequence
Classroom/ Fred Sherwood Command Team Trainer		After EO 003.02
Time	**Classification**	**Instructor**
4000 mins (1000 mins per student, 80x50 min periods)	Confidential	- Min Qualification: AILU - Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	Method
1) Executing periscope safety techniques eyes-only attack	Practical

Scenario	Students will be in the Fred Sherwood Command Team Trainer conducting CASEX A5 IAW with the SOCT Planning Guide Annex A and the NoSMCC Runs (SEPCOR). Students are expected to use the techniques in NCPM 401(B) Chapters 8 and 9 IAW to safely conduct the CASEX A5
Activity Steps	1) Students are given references and resources below 2) Students will be given a run as per CSFORD 4502-1 SOCT Planning Annex A or NoSMCC Runs (SEPCOR) and be expected to conduct the CASEX A5 in a safe manner 3) The roles students will fill are: a. Attack Periscope (Assessed) b. Local Operations Plot Officer (support role) c. Look sheet scribe (support role) 4) The instructor will build up the students from 1 ship to the goal of a 3 ship run

Resources	References
1) Fred Sherwood Command Team Trainer 2) Control Room Team (Recomened 3x Sonar OPS, 2X Fire Control Operators, 1X 2OOW) 3) MLD	1) VCSOPs, VCEOPS, VCSSOs 2) CFCD 110B 3) NCPM 401 (B) 4) NCPM 411 STOPs 5) AXP 1 6) AXP 5 7) ATP 1 8) ATP 28 9) CSOs 10) CSFORD 4502-1 SOCT Planning Guide

Notes to Instructor

During a 50 min period each student should expect to conduct 1 run as the Attack periscope (assuming 4 students) with other students filling in roles as required. At the end of the 4000min allotted time each student should have completed 80 runs (80x50min periods)

The standard at the end of the allotted time should be to conduct a 2 ship CASEX A5 with the goal of conducting a 3 ship CASEX A5.

EC's shall be conducted during the last 10 runs. Students must be able to pass one run with 2 ships.

Item	EC 003.03 – Conduct Eyes Only Attacking	SAT	UNSAT	Remarks
1. Safety				
Act within LI and proper action conducted ***		☐	☐	**Critical point**
Ship entered GDR and proper action conducted ***		☐	☐	**Critical point**
2. Periscope Skills				
Understand Situational Awareness		☐	☐	
Conduct periscope math		☐	☐	
Use masts effectively to avoid counter detection (Mast Exposure)		☐	☐	
Conduct Standard Drills & Reports IAW CANSADS & NCPM 401		☐	☐	
Understand Time Management		☐	☐	

Score: **/ 7**

Trainee must achieve 6 / 7 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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EO 003.04A – Run a Force Generation Operation

Conditions		Sequence
Classroom		After EO 003.03
Time	**Classification**	**Instructor**
400 mins	Confidential	- Min Qualification: AILU - Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer - Submarine Senior Sonar Operator with Sonar Watch Leader qualification

Learning Objectives	Method
1) Advising Command on UWW Safety rules to include: a. Underwater communications 2) Advising Command on Anti-Surface Warfare Safety 3) Direct tactical maneuvering to include; a. Applying towed array tactics; and b. Implement Anti-surface warfare tactics c. Implementing UWW Tactics 4) Managing UWW Operations 5) Interpreting meteorological oceanographic and acoustic environmental info 6) Managing Anti-Surface warfare Operations	Case Study

Scenario	Students will be given assigned to write briefing notes on each of the learning objectives. They will be given the references below the instructors will be available for consultation. The aim will be to write the briefing notes in a format that will aid the students during the last simulator phase and during their tours as SMOROs. In a way this should help to serve as a quick reference package
Activity Steps	1) Students are given references and resources below 2) Students will are required to write a briefing note on each of the learning objectives. The depth of knowledge will be dictated by the instructor but at t minimum are expected to cover the knowledge that is expected of OTT2/WST2/AMSRTs. Therefore the level of knowledge should include: a. CASEX B series with MPAs and MH b. CASEX A and C series with surface units c. CASEX S with other submarines d. Towed Array Operations

	e. Acoustic Analysis to include oceanography and how it can be used to help fight the submarine 3) If extra time is needed it will be treated as homework

Resources	References
1) CSNI Access (each student should have their own CSNI account and access to a CSNI terminal) 2) Classified Stand Alone laptop with associated printer 3) USB Memory stick for transferring information between CSNI and laptops 4) Presentation hardware either a projector or applicable Large Screen	1) VCSOPs, VCEOPS, VCSSOs 2) CFCD 110B 3) NCPM 401 (B) 4) NCPM 411 STOPs 5) AXP 1 6) AXP 5 7) ATP 1 8) ATP 28 9) CSOs

Notes to Instructor

Students will check in with instructor about status of briefing notes and the instructor shall help them fill in the knowledge gaps so they can complete their notes. The goal of this is that the students can use the information gathered as a ready reference when actually conducting the job of a SMORO.

The EC are the briefing notes

EO 003.04B – Complete Post Watch Records

Conditions		Sequence
Classroom		
Time	**Classification**	**Instructor**
150mins	Unclassified	- Min Qualification: AILU - Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	Method
1) Compile and complete records to include but not limited to: a. Quicklook/Trackex Msg b. First Impressions Reports c. OOW Narrative d. RCN Post Deployment Report	Case Study

Scenario	Students will be given templates of above records listed in Learning Objectives and will be given a case study. Students will then be expected to produce examples using the templates
Activity Steps	1) Students are given references below 2) Students are given a case study by the instructor and templates of above records. Students then are then to produce examples of each record. 3) Upon completion students will present their examples to the instructor 4) Instructor will review examples and provide guidance on how to properly draft the records to a command level.

Resources	References
1) UNCLASS DWAN Computer with Microsoft Office (one per student) 2) Printer for UNCLASS DWAN Computer 3) Classroom	1) NAVORD 8010-1- RCN Lessons Learned Policy 2) CFCD 110B 3) Canadian Submarine Records Manual

Notes to Instructor

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- For ease of instruction and to keep the material unclassified students are to produce example reports using fake names and locations. The intent of the exercise is to go through the process of producing proper submarine records
- Completion of all records examples will constitute the EC for this EO

EO 003.04C – Complete General Operations Plot

Conditions		Sequence
Classroom		
Time	**Classification**	**Instructor**
100mins	Unclassified	- Min Qualification: AILU - Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	Method
Compile a general operations plot	Case Study Guided Discussion

Scenario	Students will be given a scenario and an ECPINS Laptop. They are they expected to produce a GOP to the quality that is of a Daily Command Brief at sea.
Activity Steps	1) Students are given references below 2) Students are given a ECPINS Laptop and a geographical area and will be briefed on traffic patterns, threats, targets, and the situational area 3) Students will be given time to produce a GOP and if more time is required this will be homework 4) Students are expected to export their ECPINS Screen captures to PowerPoint as if they were producing a Daily Command Brief 5) At a minimum the GOP should include: a. Merchant Traffic Patterns (observed frequency over the given time period, nations, cargo, NPOC) b. Threat bearing and observed OPFOR (to include Aircraft sightings and behavior) c. Target bearings d. Area of responsibility and what the submarine is supposed to conduct ISR on e. Any more details the student deems will assist command in updating situational awareness 6) Upon completion students will present their examples to the instructor 7) Instructor will review examples and provide guidance on how to properly make a GOP that is expected to go into a Daily Command Brief.

Resources	References

1) UNCLASS DWAN Computer with Microsoft Office (one per student) 2) Printer for UNCLASS DWAN Computer 3) Classroom 4) Microsoft PowerPoint 5) ECPINS Laptop with applicable charts and licenses loaded 6) Means to transfer ECPINS screen captures to DWAN computer for exporting to Microsoft Power Point	1) CFCD 110 (B) 2) Canadian Submarine Records Manual 3) NCPM 411 STOPS 4) SMPs 5) NCPM 401 (B)

Notes to Instructor

- For ease of instruction and to keep the material unclassified students are to produce the GOP using fake names and scenarios but can use real locations. The SOG or the MARLOAS IVO of the Traffic lanes is a recommend geographic area but it is not limited to that. The intent of the exercise is to go through the process of producing a proper GOP
- Completion of the GOP will constitute the EC for this EO

PC 003 – Run a Dive Watch in Support of Force Generation Operations

Scope	Conditions	Sequence
PO 003	-Fred Sherwood Submarine Trainer -unrestricted Visibility -Sea State 1 -Day	
Time	**Classification**	**Instructor**
800 mins (200 per student)	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Scenario	A simulated submarine at sea with no material defects. Students will start in an area with the expectation of making it to an RV while safely maneuver in med to high density traffic (maximum 8). Military Units may be present during these runs but will not be considered a threat at this level of the training. Students given resources as listed below. Denied assistance.
Assessment Steps	1) Student will be provided a start point with an applicable navigation objective. These runs will be very similar to what is encountered in OTT1 2) Student will be acting as a dived OOW with a full control room team. 3) Student will encounter close quarter contacts and fishing vessels 4) Students will encounter navigational hazards 5) Students will be expected to safely operate the submarine in med/high density traffic to be assessed via the checklist below 6) Instructor will assess according to checklist. 7) Feedback given by instructor after the run.

Resources	References
1) Fred Sherwood Command Team Trainer	1) STOPS 2) SOPs 3) EOPs 4) CSOs

2) Control Room Team (Recommended 3x Sonar OPS, 2X Fire Control Operators, 1X 2OOW) 3)MLD	

Notes to Instructor
Ensure students respond to all enviromentals in practice runs. To be similar to an OTT1 run. Students to be evaluated on safety as tactical considerations to be addressed in PC 004. Return to PD should be included in PC scenario.

Limitations	Instructors Aid
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

Item	PC 003 Run a Dived Watch in Support of Force Generation Operations	SAT	UNSAT	Remarks
1. Submarine Routine/State				
Conduct Plot Brief to include: - Situation/Geography - Mission - Team Input - Sensor Employment		☐	☐	
Conduct Watch/Outstation Brief		☐	☐	
2. Situational Appreciation				
Maintenance of Priorities		☐	☐	
Manoeuvre Submarine		☐	☐	
Assigning contact tracking priorities		☐	☐	
Conduct Drills and Reports IAW NCPM 401		☐	☐	
Demonstrate understanding of Submarine Manoeuvring Limitations: - MLD - Towed Array		☐	☐	
3. Tactical Appreciation				
Adopt IARL as appropriate		☐	☐	
Calculate LI		☐	☐	
React to Environment as appropriate		☐	☐	
4. Safety				
Fishing Vessel Avoidance *** - Identify - Manoeuvre IAW CFCD 110		☐	☐	**Critical**
Close Quarters Contact Situation *** - Identify - Manoeuvre IAW CSOs		☐	☐	**Critical**
Navigation *** - POE IAW CFCD 130 - WSM/PMI		☐	☐	**Critical**
Supervise SCOOW - Understand System Losses		☐	☐	

5. Periscope Skills			
Understand Situational Awareness - Contact Avoidance	☐	☐	
Conduct periscope math	☐	☐	
Display Confidence & Presence	☐	☐	
Use masts effectively to avoid counter detection (Mast Exposure)	☐	☐	
Conduct Standard Drills & Reports IAW CANSADS & NCPM 401***	☐	☐	**Critical**
Understand Time Management	☐	☐	
6. TBP Analysis			
Contact Management	☐	☐	
Initial Target Solution	☐	☐	
Use of Formulae	☐	☐	
TBP Analysis	☐	☐	
Team Coordination	☐	☐	
Interpretation of B° Curves	☐	☐	
ZIG Recognition	☐	☐	

Score: / 27

Trainee must achieve 24/27 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment.

Positive Feedback	**Improvement Feedback**

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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PO 004 – Run a Dive Watch in Support of Force Employment Operations

1. **Performance**: Performance Statement

2. **Conditions**:

   a. Given:

      a. A submarine at sea or the Fred Sherwood Command Team Trainer ;
      b. Command direction; and
      c. References.

   b. Denied: Nil

   c. Environment: At sea in any weather condition

3. **Standard**: The SMORO student shall Run a Dived Watch in support of Force Employment operations to include:

   a. Directing Submarine routines (T 0004) to include:

      a. Directing action stations;
      b. Directing attack teams;
      c. Directing quiet states;

   b. Maintaining Safety by:

      a. Interpreting the target Motion Analysis (TMA) Plots (T 0024)
      b. Directing EOPS (T 0005);
      c. Advising Command on UWW Safety (BT 0340);
      d. Advising Command on Anti-Surface Warfare Safety (BT 0330);
      e. Executing periscope safety techniques eyes-only attack (T 0026);
      f. Assigning contact tracking priorities (T 0025)

   c. Conducting Force Employment operations to include;
      a. Creating daily plan to include;
         (a) Snorting
         (b) Domestic routines
         (c) Communications
         (d) Environmentals
         (e) Events

      b. Directing tactical maneuvering (T 0009) to include;
         (a) Applying towed array tactics; and
         (b) Implementing Anti-surface warfare tactics (BT 0300);
         (c) Implementing UWW Tactics (BT 0310)

c. Interpreting meteorological oceanographic and acoustic environmental info (T 0022);

d. Directing combat information organization (T 0017) include;

(a) Directing combat system line-ups (T 0010); and

(b) Directing personal;

e. Executing submarine operations (T0014);

(a) Managing UWW Operations (BT 0110);

(b) Managing Anti-Surface warfare Operations (BT 0100);

(c) Managing ISO to include;

(a) Periphot;

(b) Sensor drop;

(c) Combat search and rescue;

(d) Agent rendezvous;

(e) Special forces operations;

(f) Underwater look;

(g) Air-sub cooperation;

(h) Support operations; and

(i) Basic Intelligence Gathering (BINT);

f.Complete post watch requirements to include;

(a) Patrol records (Post Deployment Report, FIR, OOW narrative); and

(b) General Operations Plot (GOP)

4. Training Limitations: Students will not be retested on the training points from paras 3(b) or 3(f). They will be expected to carry on the standard from PO 003 in PC 004.

EO 004.01 – Directing Submarine Routines

EO 004.01A – Understand Different Submarine Routines

Conditions		Sequence
Classroom/Campus		
Time	**Classification**	**Instructor**
50 min	UNCLAS	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	Method
1) Understand the different Submarine Routines and when to use them	Guided Discussion

Scenario	Students are to discuss and create a briefing note on Submarine Routines and when to utilize them

Activity Steps	- The students will read the appropriate references beforehand as homework.
	- The instructor will engage the students in a guided discussion on when each routine is best utilized to include:
	  o Action Stations
	  o Attack teams
	  o Quiet states
	  ▪ When and what the limitations on equipment and personnel there are.
	  o Battery Conservation State
	  ▪ What are the limitations
	  o Shut Off for Attack/Counter Attack.
	  ▪ What are the differences between them
	  o SSE State
	  ▪ What are the different states and when would you use each

Resources	References
1)Note books 2)Writing instruments 3) Master Implementation Plan Template 4) Calendar Template 5) UNCLASS DWAN Computer with Microsoft Office (one per student) 6) Printer	1) VCSSOs 2) NCPM 401 3) VCSOPs

Notes to Instructor

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- There is no EC for this EO. The PC will be covered in the practical portion by the student considering and assuming the proper routine prior to their run

EO 004.02 –Maintain Submarine Safety

004.02A – Return to Periscope Depth

Conditions		**Sequence**
Fred Sherwood Command Team Trainer		After EO 003.01
Time	**Classification**	**Instructor**
2000 mins (500 mins per student)	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	**Method**
a) Interpreting the target Motion Analysis (TMA) Plots (T 0024) b) Directing EOPS (T 0005); c) Advising Command on UWW Safety (BT 0340); d) Advising Command on Anti-Surface Warfare Safety (BT 0330); e) Executing periscope safety techniques eyes-only attack (T 0026); f) Assigning contact tracking priorities (T 0025)	Practical

Scenario	Students will be in the Fred Sherwood Command Team Trainer in the role of OOW. They will be given resources and references as listed below. Students will be given a scenario similar to an OTT2 run where they will need to return the submarine to periscope depth during an operational scenario with light to moderate contact density, and a given surface & air threat..
Activity Steps	1) Students are given references and resources below 2) Students will be given a run similar to OTT2: a. FE in while deployed on an FE patrol waters b. Light to Moderate contact density c. given OpFOR 3) The students will then be expected to return the submarine to periscope depth 4) The instructor will then give guidance and feedback

Resources	**References**

1) Fred Sherwood Command Team Trainer 2) Control Room Team (Recommended 3x Sonar OPS, 2X Fire Control Operators, 1X 2OOW) 3) MLD	1) VCSOPs, VCEOPS, VCSSOs 2) CFCD 110B 3) NCPM 401 (B) 4) NCPM 411 STOPs 5) AXP 1 6) AXP 5 7) ATP 1 8) ATP 28 9) CSOs

Notes to Instructor
This EO was covered in 003.02. The students are expected to carry on the standard from EO 003.02 with the addition of tactical considerations. If the standard is not being met remedial training in the form of more runs will be conducted.
Each student will receive at a minimum 5 runs (each run will consist of 2x50 min periods). The instructor will assist the students on a decreasing scale with heavy guidance during runs 1 & 2, light guidance during run 3 & 4, and almost no guidance during run 5 The 5th run will be the EC and the instructor will grade on a pass/fail on whether the student possess the basics of returning a submarine to periscope depth This is a critical EC and the below checklist is to be used for assessing

EO 004.03 – Create a Daily Plan

Conditions		Sequence
Classroom/Campus		After EO 004.01
Time	**Classification**	**Instructor**
100 min	UNCLAS	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	Method
Creating a Daily Plan to include: a) Snorting (to include battery charging predictions) b) Domestic routines c) Communications d) Environmentals e) Events (CASEXs, Missions, etc.)	Case Study Guided Discussion

Scenario	Students will be given a daily planner template with a scenario and are expected to create a daily planner to incorporate all the the tasks in the learning objective
Activity Steps	- The instructor will give each student a scenario and have them complete a daily planner - The instructor will provide the template and the standard that will be considered satisfactory - The instructor will also show students on how to efficiently complete a daily planner - The students will complete the daily planner as homework - The next day the instructor will grade on a pass/fail basis - Students may choose to complete daily planner by hand or electronic means - Students will have 2x50 min time blocks to complete this task (not including time for homework). The first block will be used to explain to the students on who to create a daily planner. The students will then complete the daily planner as homework. The second block will be used as review for the instructor to give feedback

Resources	References
1)Note books 2)Writing instruments 3) Daily Planner Template 4) UNCLASS DWAN Computer with Microsoft Office (one per student) 5) Printer	1) CFCD 110 SUBOPINST 2) NCPM 401 CIO ORG 3) NCPM 411 STOPS 4) CSOs

Notes to Instructor

- The EC will be the completion of this assignment and is not considered a critical EC.

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EO 004.04 – Direct Tactical Maneuvering

Conditions		Sequence
Classroom/Campus		After EO 004.03
Time	**Classification**	**Instructor**
100 min	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	Method
1) Applying towed array tactics; and 2) Implementing Anti-surface warfare tactics (BT 0300); 3) Implementing UWW Tactics (BT 0310) 4) Implementing ASW Evasion Tactics	Case Study Table Top Presentation (Fishbowl)

Scenario	Students will be given the assignment of creating a briefing note and presentation to the instructor on the learning objectives.
Activity Steps	- The instructor will provide the students with a handout in which a briefing note and table top presentation will need to be produced. - The instructor will discuss with students on which tactics need to be outlined in the briefing note. - The students will then research in the references and produce a briefing note and then prepare a table top presentation for the instructor - The instructor will then reconvene and table top with the students different tactical scenarios with the students and they are to present their solutions as researched

Resources	References
1)Note books 2)Writing instruments 3) Daily Planner Template 4) UNCLASS DWAN Computer with Microsoft Office (one per student) 5) Printer 6) CSNI Access (each student should have their own CSNI account and access to a CSNI terminal)	1) CFCD 110 SUBOPINST 2) NCPM 401 CIO ORG 3) NCPM 411 STOPS 4) CSOs

7) Classified Stand Alone laptop with associated printer 8) USB Memory stick for transferring information between CSNI and laptops 9) Presentation hardware either a projector or applicable Large Screen	

Notes to Instructor

- The EC will be the completion of this assignment and is not considered a critical EC.

EO 004.05 – Interpret meteorological oceanographic and acoustic environmental info

Conditions		Sequence
Classroom/Campus		
Time	**Classification**	**Instructor**
400 min	SECRET	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer Guest Instructor: -Senior Submarine Sonar Operator/ UC1. Min Qualification: AILE

Learning Objectives	Method
1) Interpret meteorological oceanographic and acoustic environmental information	Guided Discussion Fishbowl/ Table Top Scenarios

Scenario	- This training will primarily be provided by an AILE qualified Sonar Operator with the main instructor (AILU) inputting Command perspective as required
Activity Steps	- The guest instructor will deliver a guided discussion on interpreting meteorological oceanographic and acoustic environmental information - On completion of the guided discussion the instructor will conduct fishbowl/table top scenarios with the students to go practice information learned

Resources	References
1)Note books 2)Writing instruments 3) Whiteboard with markers 4) Projector (for SECRET presentations) 5) Laptop (for SECRET presentations)	1) CFCD 110 SUBOPINST 2) NCPM 403 3) NCPM 411

Notes to Instructor
- There will be no EC for this EO. Performance will be measured in PO 004 by the students being able to interpret information given to them during their runs

EO 004.06 – Directing combat information organization

Conditions		Sequence
Classroom/Campus		
Time	**Classification**	**Instructor**
100 min	UNCLAS	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	Method
1) Directing combat information organization to include; a. Directing combat system line-ups; and b. Directing personnel	Case Study Guided Discussion

Scenario	- The instructor and students will discuss what different combat system line ups and personnel are required
Activity Steps	- Each student is to produce a briefing note and presentations using the references below. The briefing notes will cover: o All the different personnel configurations, people required, and when would they be used o The different combat line ups and when they would be used - The students will present the briefing notes to the instructor - The instructor will conduct a guided discussion with the students on that is required and what they are missing

Resources	References
1)Note books 2)Writing instruments 3) Daily Planner Template 4) UNCLASS DWAN Computer with Microsoft Office (one per student) 5) Printer	1) CFCD 110 SUBOPINST 2) NCPM 401 CIO ORG 3) NCPM 411 STOPS 4) CSOs 5) VCSSOs

Notes to Instructor
- The EC will be the completion of the briefing note and is not considered a critical EC.

EO 004.07 – Run a Force Employment Operation

Conditions		**Sequence**
Classroom/Campus		
Time	**Classification**	**Instructor**
800 mins	UNCLAS	- Min Qualification: AILU - Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer - Submarine Senior Sonar Operator with Sonar Watch Leader qualification

Learning Objectives	**Method**
1) Managing UWW Operations (BT 0110); 2) Managing Anti-Surface warfare Operations (BT 0100); 3) Managing ISO to include; a. Periphot; b. Sensor drop; c. Combat search and rescue; d. Agent rendezvous; e. Special forces operations; f. Underwater look; g. Air-sub cooperation; h. Support operations; and i. Basic Intelligence Gathering (BINT);	Case Study

Scenario	Similar to EO 003.04, students will be assigned to write briefing notes on each of the learning objectives. They will be given the references below the instructors will be available for consultation. The aim will be to write the briefing notes in a format that will aid the students during the last simulator phase and during their tours as SMOROs. In a way this should help to serve as a quick reference package. For each ISO the students are expected to produce a generic event checklist which will aid them in the PC.
Activity Steps	- Students are given references and resources below - Students will are required to write a briefing note on each of the learning objectives. The depth of knowledge will be dictated by the instructor but at a minimum are expected to cover the knowledge that is expected of for AMSRT/MSRT. - If extra time is needed it will be treated as homework

Resources	References
1) CSNI Access (each student should have their own CSNI account and access to a CSNI terminal) 2) Classified Stand Alone laptop with associated printer 3) USB Memory stick for transferring information between CSNI and laptops 4) Presentation hardware either a projector or applicable Large Screen	1) VCSOPs, VCEOPS, VCSSOs 2) CFCD 110B 3) NCPM 401 (B) 4) NCPM 411 STOPs 5) AXP 1 6) AXP 5 7) ATP 1 8) ATP 28 9) CSOs

Notes to Instructor

Students will check in with instructor about status of briefing notes and the instructor shall help them fill in the knowledge gaps so they can complete their notes. The goal of this is that the students can use the information gathered as a ready reference when actually conducting the job of a SMORO.

The EC are the briefing notes and should be used on the PC portion of the PO 004

PC 004 – Run a Dive Watch in Support of Force Employment Operations

Scope	Conditions	Sequence
PO 004	-Fred Sherwood Submarine Trainer -Unrestricted Visibility, -Sea State 1 -Day	After PC 002, 003, 008
Time	**Classification**	**Instructor**
Developmental 1600 min (includes 2 development runs per student) Assessment 800 min (200 mins per student for assessment) Total 2400 min	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer -Recommended Guest Assessor Qualification: SMCC

Scenario	A simulated submarine at sea with possible material defects. Students will start in an area and expect to complete one of several submarine operations such as ISO, BINT, Warfare. Students are expected to experience opposition during their assessments in the form of air, surface, or sub-surface threats. Students to make effective decisions based on submarine priorities. Assessments to be conducted IAW checklist below. Students given resources as listed below. Denied assistance. Students are required to produce an Operations brief IAW current format from CSF prior to every run.
Assessment Steps	1) Student will be provided a start point with an applicable objective. These runs will be very similar to what is encountered in OTT2 2) Student will be acting as a dived OOW with a full control room team.

	3) Student will encounter close quarter contacts and fishing vessels 4) Students will encounter navigational hazards 5) Students will be expected to defend the submarine and remain undetected in a threat environment. 6) Instructor will assess based on checklist. 7) Feedback will be given on completion of the run.

Resources	**References**
1) Fred Sherwood Command Team Trainer 2) Control Room Team (Recommended 3x Sonar OPS, 2X Fire Control Operators, 1X 2OOW) 3) Personnel for Attack Teams 4) MLD	1) STOPS (NCPM 411) 2) SOPs 3) EOPs 4) CSOs 5) Duty Plan 6) Threat Pack 7) MK48 User Manual (FMS) 8) Ops Pack 9) ATP-18 and 22 10) CFCD 106 11)NCPM 401

Notes to Instructor
Ensure students respond to all enviromentals in practice runs. To be similar to an OTT2 run. Students to be evaluated on safety and tactical considerations as seen in checklist. RTPD shall be added to scenario. Each run shall take 4x50 minute periods. Students will be given at a minimum 1 developmental run and are required to pass 1 assessed run. They instructor may give the student up to 3 developmental runs and 3 assessed runs. This PC should be done normally at the end of the course as it serves as a culmination of course.

Limitations	Aide from Instructor
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

Item	PC 004 Run a Dived Watch in Support of Force Employment Operations	SAT	UNSAT	Remarks
1. Submarine Routine/State				
Assume Appropriate Manning/Machinery Configuration: - Action Stations - Attack teams - Quiet States - Battery Conservation State - Shut Off for Attack/Counter Attack - SSE State		☐	☐	
Conduct Watch/Outstation Brief		☐	☐	
2. Tactical Appreciation				
Maintenance of Priorities - Safety - Remain Undetected - Achieve the Aim		☐	☐	
Manoeuvre Submarine Tactically - IAW Duty Plan - IAW STOPs - Towed Array Tactics		☐	☐	
Manage Tactical Picture		☐	☐	
Assign contact tracking priorities to include: - Threat Management - Situational Awareness - Environmental Awareness - RMP Management		☐	☐	
Conduct Drills and Reports IAW CANSADS & NCPM 401***		☐	☐	**Critical**
Demonstrate understanding of Submarine Manoeuvring Limitations: - MLD*** - Towed Array		☐	☐	
Demonstrate Sensor Prioritisation		☐	☐	
Adopt IARL as appropriate		☐	☐	
Calculate LI		☐	☐	
React to environmentals as appropriate		☐	☐	
		☐	☐	
3. Safety				

Fishing Vessel avoidance*** - Identify - Manoeuvre IAW CFCD 110	☐	☐	**Critical**
Close Quarters Contact Situation*** - Identify - Manoeuvre IAW CSOs	☐	☐	**Critical**
Navigation*** - POE IAW CFCD 130 - WSM/PMI	☐	☐	**Critical**
Supervise SCOOW - Understand System Losses - Manage Material Limitations - Considered EOP Amendments IAW tactical scenarios	☐	☐	
4. Periscope Skills			
Understand Situational Awareness*** Contact Avoidance	☐	☐	
Conduct periscope math***	☐	☐	
Display Confidence & Presence	☐	☐	
Use masts effectively to avoid counter detection (Mast Exposure)	☐	☐	
Conduct Standard Drills & Reports IAW CANSADS & NCPM 401***	☐	☐	
Understand Time Management ***	☐	☐	
5. TBP Analysis			
Contact Management	☐	☐	
Initial Target Solution	☐	☐	
Use of Formulae	☐	☐	
TBP Analysis	☐	☐	
Team Coordination	☐	☐	
Interpretation of B° Curves	☐	☐	
ZIG Recognition	☐	☐	
6. Weapons Employment			
Conduct Proper Drills and Procedure IAW CANSADS	☐	☐	
Display Weapon Knowledge	☐	☐	
Calculate Weapon Settings / Presets	☐	☐	
Execute Weapon Control IAW CANSADS***	☐	☐	
Consider Safety Awareness***	☐	☐	

- Identify Safety Measures (ASH, Stratum)			
Orchestrate Target Co-Ord / Awareness	☐	☐	

Score: / 36

Trainee must achieve 33/36 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment.

Positive Feedback	**Improvement Feedback**

Overall Assessment – Circle One

Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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PO 005 – Advise Command on Force Protection Requirements

1. **Performance**: Advice Command on Force Protection Requirements

2. **Conditions**:

   a. Given:

      (1) References;
      (2) DWAN; and
      (3) CSNI;

   b. Denied: Nil.

   c. Environment:

      (1) Day or night; and
      (2) Alongside or at Sea.

3. **Standard**: The SMORO shall advise Command on Force Protection by:

   a. Advising Command on the Use of Force by:

      (1) Applying International Maritime Law to maritime operations;

      (2) Applying the Law of Armed conflict to maritime operations;

      (3) Applying the UN Convention and the Laws of the Sea to maritime operations;

      (4) Applying the use of force principles to maritime operations; and

      (5) Applying escalation of force IAW CAF Use of Force doctrine.

   b. Advising Command on Rules of Engagement by:

      (1) Applying the ROE process to include:

         (a) ROE Request;
         (b) ROE Authorization; and
         (c) ROE Implementation.

      (2) Assessing Self Defence and ROE situations; and

      (3) Applying ROE to maritime operations.

   c. Directing force protection (T 0018) by:

(1) Developing a FP plan;
(2) Developing a FP intentions message;

(3) Implementing a FP plan alongside;

(4) Implementing a FP plan while underway; and

(5) Applying RCN and Allied FP policies and doctrine.

3. **Training Limitations**: Nil.

EO 005.01 – Advise Command on CAF Use of Force

Conditions		Sequence
Classroom/Campus		
Time	**Classification**	**Instructor**
200 mins	Confidential	Guest Lecturer: JAG Qualified Legal Officer

Learning Objectives	Method
1)Applying International Maritime Law to maritime operations; 2)Applying the Law of Armed conflict to maritime operations; 3)Applying the UN Convention and the Laws of the Sea to maritime operations; 4)Applying the use of force principles to maritime operations; and 5) Applying escalation of force IAW CAF Use of Force doctrine.	Guided Discussion

Scenario	Guest lecturer will guided student through a discussion of learning objectives and how it pertains as to them as a SMORO
Activity Steps	1) The Guest lecturer will be given resources as listed below in order to present students with the relevant information required.

Resources	References
1) CSNI Access (each student should have their own CSNI account and access to a CSNI terminal) 2) Classified Stand Alone laptop with associated printer 3) USB Memory stick for transferring information between CSNI and laptops 4) Presentation hardware either a projector or applicable Large Screen 5) Access to CSNI Trinity Resources (each student should have their own CSNI account and access to a CSNI terminal) 6) Force Protection Brief Template	1) NAVORD 2100-1 RCN Force Protection 2) Threat Assessment on relevant port 3) B-GJ-005-501/FP-001 Use of Force in CF Operations – Aug 2008 4) Naval Force Protection SOPS version 4

Notes to Instructor

- Guest lecturer must be a qualified JAG Legal officer.

EO 005.02 – Advising Command on Rules of Engagement

Conditions		Sequence
Classroom/Campus		
Time	**Classification**	**Instructor**
100 mins	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	Method
Applying the ROE process to include: (a) ROE Request; (b) ROE Authorization; and (c) ROE Implementation.	Guided Discussion Case Study

Scenario	Students will be given a scenario where they will be required to submit an ROE Request, receive and ROE Authorization, and conduct ROE Implementation
Activity Steps	1) The instructor will assign each student with a case study in which the student will be required to submit an ROE request. 2) The instructor will guide the student in how to properly submit a ROE Request 3) Student will submit the ROE Request to the instructor and then receive a ROE Authorization 4) Students will then take the ROE Authorization and conduct ROE Implementation including drafting a ROE Implementation message.

Resources	References
1) CSNI Access (each student should have their own CSNI account and access to a CSNI terminal) 2) Classified Stand Alone laptop with associated printer 3) USB Memory stick for transferring information between CSNI and laptops 4) Presentation hardware either a projector or applicable Large Screen 5) Access to CSNI Trinity Resources (each student should have their own CSNI account and access to a CSNI terminal)	1) NAVORD 2100-1 RCN Force Protection 2) Threat Assessment on relevant port 3) B-GJ-005-501/FP-001 Use of Force in CF Operations – Aug 2008 4) Naval Force Protection SOPS version 4 5) MC 362/1 NATO Rules of Engagement

6) Example ROE request, ROE Authorization, and ROE Implementation messages	

Notes to Instructor

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EO 005.03 – Coordinating a Force Protection Plan

Conditions		Sequence
Classroom/Campus		
Time	**Classification**	**Instructor**
100 mins	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	Method
(1) Developing a FP plan; (2) Developing a FP intentions message; (3) Implementing a FP plan alongside; (4) Implementing a FP plan while underway; and (5) Applying RCN and Allied FP policies and doctrine.	Guided Discussion Case Study

Scenario	Students will be given a scenario where they will be required to submit a FP intentions message and implement a FP Plan
Activity Steps	1) The instructor will assign each student with a case study in which the student will be required to submit an FP Intentions message 2) The instructor will guide the student in how to properly submit a FP Intentions message 3) Student will submit the FP Intentions message to the instructor 4) Students will then take the implement their FP plan based off the FP Intentions message 5) Students will produce a PowerPoint presentation on their FP implementation plan based on the scenario

Resources	References
1) CSNI Access (each student should have their own CSNI account and access to a CSNI terminal) 2) Classified Stand Alone laptop with associated printer 3) USB Memory stick for transferring information between CSNI and laptops 4) Presentation hardware either a projector or applicable Large Screen 5) Access to CSNI Trinity Resources (each student should have their own CSNI account and access to a CSNI terminal)	1) NAVORD 2100-1 RCN Force Protection 2) Threat Assessment on relevant port 3) B-GJ-005-501/FP-001 Use of Force in CF Operations – Aug 2008 4) Naval Force Protection SOPS version 4 5) MC 362/1 NATO Rules of Engagement

6) Example ROE request, ROE Authorization, and ROE Implementation messages	

Notes to Instructor

- Students are to submit a FP intentions message and then produce a PowerPoint presentation on their FP implementation plan based on the scenario. This will serve as the EC

PC 005 - Advise Command on Force Protection Requirements

Scope	Conditions	Sequence
PO 005	Classroom	After lesson PO 005.XX
Time	**Classification**	**Instructor**
200 min (100 mins for preparation 100 mins for briefing)	Confidential	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Scenario	The student is acting as the submarine's Operation Officer in preparation for a berthing in an away from home port scenario which requires Force Protection to be considered. (Each student will have a different scenario) The student will be required to produce a Force Protection plan which includes: - The Force Protection brief to the crew - The Force Protection Plan message
Assessment Steps	1) The student is given resources as listed below and they are allowed to seek help from external resources 2) The student is expected to know how to request (RFI) a Threat Assessment from Trinity/N2 via the CSNI website. The student is also expected to produce a Force Protection Plan message as well as a brief 3) The instructor is expected to provide all resources and references as listed below. The instructor will assign the port for each individual student and if possible make contact with Trinity/N2 to allow the students to submit their own RFIs for the Threat Assessments. If that is not possible the instructor is to provide an old threat assessment to the students. 4) The instructor will assess this PC via checklist listed below 5) After the assessment the instructor will have a feedback session with the students and allow them to keep soft copies of their deliverables

Resources	References
1) CSNI Access (each student should have their own CSNI account and access to a CSNI terminal) 2) Classified Stand Alone laptop with associated printer	1) NAVORD 2100-1 RCN Force Protection 2) Threat Assessment on relevant port 3) B-GJ-005-501/FP-001 Use of Force in CF Operations – Aug 2008 4) Naval Force Protection SOPS version 4 5) MC 362/1 NATO Rules of Engagement

3) USB Memory stick for transferring information between CSNI and laptops 4) Presentation hardware either a projector or applicable Large Screen 5) Access to CSNI Trinity Resources (each student should have their own CSNI account and access to a CSNI terminal) 6) Force Protection Brief Template	

Notes to Instructor

- This PC is divided into two parts. The first part is a Force Protection Plan message as indicated in NAVORD 2100-1. The second part is a brief in which the intended audience would be to all members of the duty watch in the intended foreign port.

Limitations	N/A The student is encouraged to seek out as many resources and ways to conduct this task as possible
Impact of Failure	If the student does not satisfactorily complete the PC they will be given another Threat Assessment and will repeat it.

Item	PC 005 Advise Command on Force Protection Requirements	SAT	UNSAT	Remarks
1. Deliver Force Protection Brief				
Given Threat Assessment identify Threat Level In Port		☐	☐	
Brief team on FP Requirements based on level		☐	☐	
Brief team on Use of Force and Self Defence as per CF Use of Force Manual		☐	☐	
Brief team on Underway Force Protection requirements		☐	☐	
2. Draft Force Protection Plan Message				
Identify the Force Protection level the Submarine Intends to use		☐	☐	
Identify Force Protection measures will be in place		☐	☐	
Identify Force Protection amendments and justifications		☐	☐	

Score: **/ 7**

The trainee must achieve 7/7 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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PO 006 – Respond to a Submarine Emergency as the Incident Controller

1. **Performance**: Performance Statement

2. **Conditions**:

   a. Given:

      a. A submarine at sea or MCT/SCT Trainer;
      b. Command direction; and
      c. References.

   b. Denied: Nil

   c. Environment: at sea in any weather condition

3. **Standard**: The member shall respond to a submarine emergency as the Incident Controller by:

   a. Proceeding to HQ1 on alarm:

   b. Directing the Damage Control (DC) organization (T 0006) by:

      a. Establishing Control;
      b. Managing SMEs; and
      c. Making informative pipes;

   c. Identifying incorrect/abnormal conditions (T00:

   d. Advising command.

4. **Training Limitations**: Nil

EO 006.01 - Respond to a Submarine Emergency as the Incident Controller

Conditions		Sequence
- Classroom (Table Top Scenario), or MCT/SCT Trainer		
Time	**Classification**	**Instructor**
400 min	UNCLAS	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or MSEO-SUB with Post HOD experience. The Ideal instructor for this PC would be the EO of Sea Training Submarines as they are considered the standard for DC in the RCN.

Learning Objectives	Method
1) Proceeding to HQ1 on alarm: 2) Directing the Damage Control (DC) organization by: a. Establishing Control; b. Managing SMEs; and c. Making informative pipes; 3) Identifying incorrect/abnormal conditions 4) Advising command.	Guided discussion Case Study Table Top

Scenario	The student is to act as the Incident Controller in a Damage Control scenario.
Activity Steps	- The instructor will conduct a guided discussion and table top exercise with the students - The instructor will run the students through an EOP from beginning to end from the role of the DCHQ1 Incident Controller - Students may be provided help from a team consisting of pers simulating an at sea environment as per VCSSO

Resources	References
1) Victoria-Class HQ1 Team as per VCSSOs 2) Classroom 3) MCT/SCT Trainer 4) Damage Control Documentation and Resources for HQ1 as listed in the	1) VCEOP 2) VCSOP 3) VCSSOS 4) TABS 5) Incident Controller Crib Template

Submarine Work Up Guided Annex Q para 1.	6) Damage Control Documentation and Resources for HQ1 as listed in the Submarine Work Up Guided Annex Q para 1.

Notes to Instructor

Difficulty should be at the level of BSSRT/WUPS PH3. This PC serves as an introduction to the student in being an IC in HQ1 and takes into account that they may not be doing the job until post OPSO

PC 006 - Respond to a Submarine Emergency as the Incident Controller

Scope	Conditions	Sequence
PO 006	- Classroom (Table Top Scenario), or - MCT/SCT Trainer	
Time	**Classification**	**Instructor**
200 min	UNCLAS	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or MSEO-SUB with Post HOD experience. The Ideal instructor for this PC would be the EO of Sea Training Submarines as they are considered the standard for DC in the RCN.

Scenario	The student is to act as the Incident Controller in a Damage Control scenario. This PC can be run in the MCT/SCT Trainer or in a table top scenario. The student is presented a DC problem by the instructor and they are to act as if they are the Incident Controller. The instructor will act as the outstations if none are available. The student as Incident Controller should be able to take the DC problem until completion and the submarine is back in a safe state. Each student will be given 50 min for assessment.
Assessment Steps	1) The student is given all the references and resources listed below. They are not to be provided their scenarios beforehand. If possible the student should be provided with relevant SMEs as listed in VCSSOs. If not the Instructor will act in those roles. 2) The Student is expected to act as the Incident Controller of a DC problem and to take the problem from turnover with the SCOOW until completion. The assessment criteria checklist is listed below 3) The instructor is expected to administer a scenario that is similar in difficulty to BSSRT/WUPS PH 3. The instructor may act as the outstations if SMEs are unavailable

	4) The assessment criteria are laid out in the checklist below

Resources	References
1) Victoria-Class HQ1 Team as per VCSSOs 2) Classroom 3) MCT/SCT Trainer 4) Damage Control Documentation and Resources for HQ1 as listed in the Submarine Work Up Guided Annex Q para 1.	1) VCEOP 2) VCSOP 3) VCSSOS 4) TABS 5) Incident Controller Crib Template 6) Damage Control Documentation and Resources for HQ1 as listed in the Submarine Work Up Guided Annex Q para 1.

Notes to Instructor

- Difficulty should be at the level of BSSRT/WUPS PH3. This PC serves as an introduction to the student in being an IC in HQ1 and takes into account that they may not be doing the job until post OPSO

Limitations	Nil
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

Item	EC / PC 006 Respond to a Submarine Emergency as the Incident Controller	SAT	UNSAT	Remarks
1. Directing the Damage Control (DC) organization				
Establish Command Priority		☐	☐	
Conduct Turnover with SCOOW		☐	☐	
Conduct SITREP Pipe		☐	☐	
Manage SMEs		☐	☐	
Develop Plan		☐	☐	
Brief Command		☐	☐	
Execute Plan		☐	☐	
Make Informative Pipes		☐	☐	
2. Walk back from Incident				
Turnover systems back to Command		☐	☐	
Coordinate Repairs		☐	☐	
Conduct Smoke Clearance (if applicable)		☐	☐	
Fall out from Incident		☐	☐	

Score: / 12

The trainee must achieve 10/12 SATs to pass this EC.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor I have discussed the results of this assessment with the candidate.	**Candidate** I have been debriefed on the result of this assessment

Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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PO 007 – Conduct Submarine Seamanship Evolutions

1. **Performance**: Performance Statement

2. **Conditions**:

   a. Given:

      a. Submarine at sea or classroom;
      b. DWAN Computer access; and
      c. References.

   b. Denied:

      a. Nil

   Environment: At sea or alongside in any weather condition

3. **Standard**: The XO shall conduct submarine seamanship evolutions to include:

   a. Berthing/Slipping by:

   b. Helicopter hoist transfer:

   c. Towed array deployment by:

   d. Boat transfer by:

   e. Towing

4. **Training Limitations**: No reasonable ability to practically assess.

EO 007.01A - Conduct Submarine Seamanship Evolutions

Conditions		Sequence
Classroom with means to conduct PPT presentations		
Time	**Classification**	**Instructor**
200 mins	UNCLAS	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Learning Objectives	Method
Conduct submarine seamanship evolutions to include: a. Berthing/Slipping b. Helicopter hoist transfer c. Towed array deployment d. Boat transfer e. Towing f. Embark/disembark SOF?	Case Study Guided Discussion

Scenario	The students will be given the references and resources listed below. Each student will be assigned a case study of one of the listed seamanship evolutions listed above. The students are then to take the case study and make a briefing note on how to do the evolution from beginning to end as their role as the XO (safety officer). Students will then have a guided discussion with the instructor on their briefing note. All Seamanship evolutions are to be written up as a briefing note. If there are not enough students to give one case study each then the instructor will assign multiple to students and if there is not enough in class time to work on it students are to do it as homework.
Activity Steps	1) Students get assigned case study 2) Students take case study and prepare briefing note on how to conduct evolution to include: a. Members required: a. Qualifications of members b. What parties and teams required to be closed up b. Equipment required to include: a. Pre evolution equipment installations if required b. Special requirements c. Sequence of Events for Evolution to include: a. Closing up times

	b. Water tight integrity c. Path for personnel d. Communications plan e. Success criteria f. Lists no go criteria: a. Navigation Safety b. Personnel c. Equipment d. Environmental 3) Students will then present their briefing note to the instructor and instructor will have a guided conversation with all students about it. 4) This is to be repeated until all evolutions are covered 5) Student will then use their briefing notes to help them with the PC

Resources	References
1) UNCLASS DWAN Computer with Microsoft Office (one per student) 2) Printer	1) VCSOPS 2) CFCD 105 3) SUBMARINE TOWED ARRAY HANDLING MANUAL (A-PD-055-003/PQ-001: SUBMARINE TOWED ARRAY HANDLING)

Notes to Instructor

- Focus is on creating the briefing notes and having guided discussion

PC 007 - Conduct Submarine Seamanship Evolutions

Scope	Conditions	Sequence
PO 007	Classroom with means to conduct PPT presentations	After Lessons in PO 007
Time	**Classification**	**Instructor**
100 mins	UNCLAS	-Min Qualification: AILU -Recommended Qualification: NWO-SUB with Post XO experience or a Submarine Commanding Officer

Scenario	The student will be acting as the sub XO and will be presented with one of the seamanship evolutions as listed in the PO. The student will be required to produce and deliver a brief with the key points as listed in the assessment checklist. Time allotted for each brief is 25 minutes per person (Max 15 min brief, 10 min questions and feedback). Students will present brief with fellow students present. Fellow students will not ask question or give comments.
Assessment Steps	1) The student is given all references and resources given below. They are allowed to seek help for external organizations as well 2) The student is expected to produce a PowerPoint presentation with the target audience being the applicable members of the submarine that will be undertaking the evolution. The brief is required to cover the points listed in the assessment criteria checklist below 3) The instructor is to provide all the resources and references below as well as assign each student one of the evolutions listed in PO 007 4) The student will present a brief on the evolution acting as the XO. This brief is meant to simulate the brief prior to closing up if they were at sea. The brief will cover criteria as listed in the checklist below 5) Students will be given feedback and be allowed to keep soft copies of briefs after the assessment.

Resources	References
1) UNCLASS DWAN Computer with Microsoft Office (one per student) 2) Printer	1) VCSOPs 2) CFCD 105

	3) SUBMARINE TOWED ARRAY HANDLING MANUAL (A-PD-055-003/PQ-001: SUBMARINE TOWED ARRAY HANDLING)

Notes to Instructor

- Assign each student one of the listed evolutions in the PO
- Students will prepare and deliver a brief as if they were the XO of the submarine in charge of safety. The target audience would be Command and the relevant parties IAW the Watch and Station Bill

Limitations	N/A The student is encouraged to seek out as many resources and ways to conduct this task as possible
Impact of Failure	Remedial Training as required and supplementary PC. Failure of supplementary PC may initiate TRB as per NPTORD 4500-54.

Item	EC / PC 007 Conduct Submarine Seamanship Evolutions	SAT	UNSAT	Remarks
1. Deliver Seamanship Brief covering one of the listed Evolutions				
Lists members required: - Qualifications of members - What parties and teams required to be closed up		☐	☐	
Lists equipment required to include: - Pre evolution equipment installations if required - Special requirements		☐	☐	
Discusses Sequence of Events for Evolution to include: - Closing up times - Water tight integrity - Path for personnel		☐	☐	
Lists out communications plan		☐	☐	
Lists success criteria		☐	☐	
Lists no go criteria: - Navigation Safety - Personnel - Equipment - Environmental		☐	☐	

Score: / 6

The trainee must achieve 5/6 SATs to pass this assessment.

Positive Feedback	Improvement Feedback

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Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor I have discussed the results of this assessment with the candidate.		**Candidate** I have been debriefed on the result of this assessment	
Assessor	Name:	Date:	Signature:
Candidate	Name:	Date:	Signature:

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PO 008 – OPERATE MULTIPLE SUBMARINE LAUNCHED WEAPONS

1. **Performance**: Operate Multiple Submarine Launched Weapons
2. **Conditions:**
   a. Given:

      (1) Submarine,
      (2) Watch teams,
      (3) References, and
      (4) Supervision.

   b. Denied: Nil; and

   c. Environment:
      (1) Onboard submarine, and
      (2) Watch dived.

3. **Standard**: The WCCO shall operate multiple submarine launched weapons (T0012) during attack by:
   a. Calculating weapon pre-sets IAW CFCD 106 and WAPP:
      (1) Environment;
      (2) Threats; and
      (3) Contact density.

   b. Conducting closing-up checks (Task number) on WCC by:
      (1) Receiving report from WSC;
      (2) Selecting WAC configuration;
      (3) Warming up weapon; and
      (4) Bringing weapon system to action state.

   c. Bringing weapon to firing point by:
      (1) Bringing up WCD page;
      (2) Selecting pre-calculated tactics;
      (3) Applying weapon pre-sets as directed; and
      (4) Reporting to AC ready to fire IAW CANSADs.

   d. Firing multiple weapons to include:
      (1) Responding to misfire or hang fire;
      (2) Conducting post launch commentary;
      (3) Salvo tactics;
      (4) Controlling multiple weapons during the pre-enable phase as required;
      (5) Controlling multiple weapons during post-enable phase as required; and

(6) Completing engagement IAW CANSAD

4. **Training Limitations**:

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EO 008.01 – CALCULATE WEAPON PRE-SETS
008.01.01 – CALCULATE WEAPON PRE-SETS

Location Conditions		**Instructional Sequence**
Classroom		Start of PO 008
Time	**Classification**	**Instructor**
300 min	Secret	Qual: AILU
Learning Objectives		**Method(s)**
1) Calculate Weapon pre-sets a. Environment b. Threats c. Contact density		• Interactive lectures

Scenario	Watch dived at SD
Activity Steps	***Introduction (5 min)*** Instructor will introduce the topic. ***Brainstorm (15 min)*** Instructor will ask class what are the pre-sets for the weapon are and have the students write them on a whiteboard. ***Lecture (80 min)*** A lecture introducing the functionality of the WAPP will be given to the students. Instructor will first provide in-depth lecture on weapon pre-sets and acoustic modes and their employment in weapon tactics. ***EDI (110)*** *Demo (30 min)* The instructor will display the WAPP 48 on a large screen in the classroom so all students can watch as the program is used to calculate single surface ship engagement at long range. Each student should have a copy of the employment manual as well to see the appropriate settings for each type of engagement. The student and instructor will calculate single surface ship engagement at short range. The WAPP 48 program will be the tool used with MK48 7AT. *Practice (80 min)*

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Following the interactive lecture students will be given surface and subsurface targets scenarios to calculate the appropriate tactics for the weapon for deep water and shallow water trainer environmentals. Environmentals to include SS, wind velocity, and historical bathy data.

The environmentals should be the same as the ones that they will see in the trainer for the practical portion of the module. Once corrected and discussed, these will be the tactics that will be used in the trainer for the remainder of the training.

Students will calculate the following calculations:

- single ship surface engagements at long and short range (already calculated as a class);
- single submarine engagements at short and long range;
- shallow water surface engagements against single and multiple targets;
- shallow water submarine engagements;
- attacks on very large ships;
- attacks on dead in the water targets/AMO attacks;
- attacks on task groups, with the instructor indicating which ship is the target;
- simultaneous enable and simultaneous detonation attacks on multiple targets; and
- firing in the vicinity navigation obstacles.

Guided Discussion (80 min)

After students finish calculations they will be reviewed as a class with the instructor. The ray traces should be discussed and evaluated, both in terms of how to read them, and how they depict sonar performance. The tactics should be recorded in both the standard format as can be entered in the WTD in SFCS, as well as in a presentation that can be used to brief a CO onboard a submarine. To include:

- discriminating a quiet target near a loud target;
- attacking a deep target below a shallow target; and
- low Doppler/high Doppler contacts, etc.

APLT can be used to examine the run characteristics of the chosen presets and a white board should be used to tabletop scenarios that cannot be replicated in the trainer. A discussion of 7AT best practices should complete the lecture.

Review / Three Star (10 min)

	Instructor will get the students to write down their three main takeaways from the lesson. Have some read out to the class. Ensure the do's and don'ts from CANSADs are covered.

Instructor Resources	References
1) Secret Hard Drive with reference material 2) Secret Laptop 3) WAPP 4) APLT 5) PowerPoint 6) White Board	1) CANSADs (NCPM 412) 2) SUBWOPs 3) CFCD 106

Notes to Instructor
• If finish lecture early proceed with EO 008.02 to maximize trainer time.

EO 008.02 –FIRE MULTIPLE WEAPONS
008.02.01 – WEAPON FIRING PRACTICAL

Location Conditions		**Instructional Sequence**
Fred Sherwood Trainer or Submarine alongside		After LG 008.01.01
Time	**Classification**	**Instructor**
400 min	Secret	Min Qual: AILU Recommended Qual: NWO-SUB with at sea XO experience or submarine Commanding Officer
Learning Objectives		**Method(s)**
1) Respond to Misfire/Hangfire 2) Post Launch Commentary 3) Salvo tactics 4) Controlling Weapons a. Pre-enable b. Post-enable 5) Complete engagement		• EDI
Scenario	Watched Dived at SD	
Activity Steps	***Introduction (5 min)*** Instructor will describe a scenario in which this performance would be done onboard a submarine and go over how the class will be run (5 min) ***EDI (380 min)*** *Demonstration (30 min)* Students will be provided with a demonstration on Mk48 Mod 7AT bring the weapon system to the action state, firing a weapon, carrying out a hangfire drill, then conducting post-launch commentary and controlling pre- and post-enable to the point of a hit. 1) Instructor will demo weapon firing slow time with explanation, and 2) Instructor will demo weapon firing real time to demonstrate standard. *Practice (350 min)* Students will have multiple practical runs to practice: 1) Respond to Misfire/Hangfire; 2) Conduct post launch commentary; 3) Control weapon during pre-enable phase; and, 4) Control weapon during post-enable phase.	

	Instructor will provide supervision and feedback. Students will watch each other's runs for training benefits. ***Review (15 min)*** Instructor will: 1) Review any common difficulties from the runs; 2) Have each student state their one main takeaway from this practice; and 3) Ask students if they have any further questions.

Instructor Resources	**References**
1) Secret Hard Drive with reference material 2) Secret Laptop 3) WAPP 4) SFCS with the ability to inject contacts	1) CANSADs (NCPM 412) 2) SUBWOPs 3) CFCD 106

Notes to Instructor
• Nil

EC 008 – OPERATE MULTIPLE SUBMARINE LAUNCHED WEAPONS

Scope	Location Conditions	Instructional Sequence
EO 008	Fred Sherwood Trainer or Submarine alongside	After LG 008.02.01
Time	**Classification**	**Instructor**
400 min	Secret	Min Qual: AILU Recommended Qual: NWO-SUB with at sea XO experience or submarine Commanding Officer

Scenario	Based off scenario pack.
Assessment Steps	**Input** – Given operational fire control system with ability to inject simulated contacts and conduct synthetic weapons firings. WAPP, CANSADs, CFCD 106, and student threat pack. Instructor will provide a scenario brief to students prior to commencing run encompassing weapon's system status and run mission. On completion of brief trainer staff will start predetermined training run. **Performance** – Each student will independently conduct weapon firings as a WCCO in a tactical environment against multiple hostile surface or subsurface contacts. Submarine will remain deep for assessed run, the control room team will be provided with accurate TMA solutions on the contacts by trainer staff in order to not impede operation of the weapon due to poor TMA. Trainee will have 90 min to complete the run. Instructor will act as the AC following correct drills and procedures IAW CANSADS, SUBWOPS, and CCS 876 drills. a. manage the CIO, provide quality control to target solutions; b. maintain safety of the submarine WRT the weapon; and c. follow and enforce drills described in CANSADS. WCCO (Student) will provide tactical recommendations to the AC based on the development of the scenario and properly execute WG commands as approved or dictated by the AC IAW CANSADs. Instructor will assess based on EC 008 checklist. **Output** – Trainee must achieve 25/28 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment. **Feedback** –Verbal debrief of EC checklist. (10 min per student after run)

Resources	References
1) WAPP, 2) CANSADs, 3) CFCD 106, and 4) student threat pack 5) Submarine Fire Control System	1) CANSADS, 2) SUBWOPS, and 3) CFCD 106
Notes to Instructor	
Students should be given EC checklist to review prior to EC.	
Limitations	Nil
Impact of Failure	Review and retest. If unsuccessful on retest continue to PC

EC 008 OPERATE MULTIPLE SUBMARINE LAUNCHED WEAPONS			
Item	**SAT**	**UNSAT**	**Remarks**
Conduct closing-up checks			
Selecting WAC configuration	☐	☐	
Request tube in remote	☐	☐	
Warm up weapon	☐	☐	
Bring weapon system to action state	☐	☐	
Warn CR for outboard movements	☐	☐	
Bring weapon to firing point			
Select tactics as directed	☐	☐	
Apply pre-sets as directed	☐	☐	
Select tube	☐	☐	
Exercise the weapon IAW CANSADs	☐	☐	
Identify weapon safety	☐	☐	critical
Report to AC ready to fire IAW CANSADs	☐	☐	
Firing Weapons			
Maintain positive control of weapon system	☐	☐	critical
Fire weapons when directed			
Salvo firing	☐	☐	critical
Target discrimination	☐	☐	critical
Respond to misfire or hang fire as required	☐	☐	
Conduct post launch commentary (war cries) IAW CANSADs	☐	☐	
Controlling weapons post launch			
Identify changes to enabling criteria	☐	☐	
Apply wire guidance commands as directed to include:			
Steer weapon	☐	☐	
Manipulate safety	☐	☐	
Change post launch pre-sets	☐	☐	
Pre-enable weapon commentary	☐	☐	
Post-enable weapon commentary	☐	☐	
React countermeasures and false homing	☐	☐	
Completing engagement			
Command shutdown weapon	☐	☐	critical

Render weapons system safe	☐	☐	**critical**
Deselect target (Complete the engagement)	☐	☐	
Select drain on tube order page	☐	☐	
Order WSC take tube in local	☐	☐	

Score: /28

Trainee must achieve 25/28 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor **I have discussed the results of this assessment with the trainee.**	**Trainee** **I have been debriefed on the result of this assessment.**

Assessor	**Name:**	**Date:**	**Signature:**
Trainee	**Name:**	**Date:**	**Signature:**

PROTECTED B (when complete

NTDC(A) Approval – 09/02/2022
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PC 008 – OPERATE MULTIPLE SUBMARINE LAUNCHED WEAPONS

Scope	**Location Conditions**	**Instructional Sequence**
PO 008	Fred Sherwood Trainer or Submarine alongside	After EC 008
Time	**Classification**	**Instructor**
400 min	Secret	Min Qual: AILU Recommended Qual: NWO-SUB with at sea XO experience or submarine Commanding Officer

Scenario	Based off scenario pack.
Assessment Steps	**Input** – Given operational fire control system with ability to inject simulated contacts and conduct synthetic weapons firings. WAPP, CANSADs, CFCD 106, and student threat pack. Instructor will provide a scenario brief to students prior to commencing run encompassing weapon's system status and run mission. On completion of brief trainer staff will start predetermined training run. **Performance** – Each student will independently conduct weapon firings as a WCCO in a tactical environment against multiple hostile surface or subsurface contacts. Submarine will remain deep for assessed run, the control room team will be provided with accurate TMA solutions on the contacts by trainer staff in order to not impede operation of the weapon due to poor TMA. Trainee will have 90 min to complete the run. Instructor will act as the AC following correct drills and procedures IAW CANSADS, SUBWOPS, and CCS 876 drills. a. manage the CIO, provide quality control to target solutions; b. maintain safety of the submarine WRT the weapon; and c. follow and enforce drills described in CANSADS. WCCO (Student) will provide tactical recommendations to the AC based on the development of the scenario and properly execute WG commands as approved or dictated by the AC IAW CANSADs. Instructor will assess based on PC 008 checklist. **Output** – Trainee must achieve 25/28 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment. **Feedback** –Verbal debrief of PC checklist. (10 min per student after run)

Resources	References
1) WAPP, 2) CANSADs, 3) CFCD 106, and 4) student threat pack 5) Submarine Fire Control System 6) PC Checklist	1) CANSADS, 2) SUBWOPS, and 3) CFCD 106
Notes to Instructor	
Students should be given PC checklist to review prior to the PC.	
Limitations	Nil
Impact of Failure	IAW NPTORD 4500-54

PC 008 OPERATE MULTIPLE SUBMARINE LAUNCHED WEAPONS			
Item	**SAT**	**UNSAT**	**Remarks**
Conduct closing-up checks			
Selecting WAC configuration	☐	☐	
Request tube in remote	☐	☐	
Warm up weapon	☐	☐	
Bring weapon system to action state	☐	☐	
Warn CR for outboard movements	☐	☐	
Bring weapon to firing point			
Select tactics as directed	☐	☐	
Apply pre-sets as directed	☐	☐	
Select tube	☐	☐	
Exercise the weapon IAW CANSADs	☐	☐	
Identify weapon safety	☐	☐	**critical**
Report to AC ready to fire IAW CANSADs	☐	☐	
Firing Weapons			
Maintain positive control of weapon system	☐	☐	**critical**
Fire weapons when directed			
Salvo firing	☐	☐	**critical**
Target discrimination	☐	☐	**critical**
Respond to misfire or hang fire as required	☐	☐	
Conduct post launch commentary (war cries) IAW CANSADs	☐	☐	
Controlling weapons post launch			
Identify changes to enabling criteria	☐	☐	
Apply wire guidance commands as directed to include:			
Steer weapon	☐	☐	
Manipulate safety			
Change post launch pre-sets	☐	☐	
Pre-enable weapon commentary	☐	☐	
Post-enable weapon commentary	☐	☐	
React countermeasures and false homing	☐	☐	
Completing engagement			
Command shutdown weapon	☐	☐	**critical**

Render weapons system safe	☐	☐	**critical**
Deselect target (Complete the engagement)	☐	☐	
Select drain on tube order page	☐	☐	
Order WSC take tube in local	☐	☐	

Score: /

Trainee must achieve 25/28 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One	
Met Standard	**Did Not Meet Standard**
Assessor **I have discussed the results of this assessment with the trainee.**	**Trainee** **I have been debriefed on the result of this assessment.**

Assessor	**Name:**	**Date:**	**Signature:**
Trainee	**Name:**	**Date:**	**Signature:**

PROTECTED B (when complete)

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PO 009 - DIRECT MULTIPLE SUBMARINE LAUNCHED WEAPONS

5. **Performance**: Direct Multiple Submarine Launched Weapons

6. **Conditions**:

a. Given:
(1) Submarine,
(2) Watch teams,
(3) References, and
(4) Supervision.

b. Denied: Nil; and

c. Environment:
(1) Onboard submarine, and
(2) Watch dived.

7. **Standard:** The AC shall direct multiple submarine launched weapons (T0012) during attack by:

a. Positioning submarine for attack;

b. Evaluating weapon pre-set calculations IAW CFCD 106 and WAPP;
(1) Environment,
(2) Threats, and
(3) Contact density.

c. Supervise closing-up checks (Task number) on WCC by:
(1) Directing the WCCO to warm up weapon; and
(2) Directing the WCCO to bring weapon system to action state.

d. Ordering tactics (Task Number) to be:
(1) selected, and
(2) Applied.

e. Verifying selected tactic for the following:
(1) Environmentals including water depth;
(2) Target range;
(3) Doppler;
(4) Alertment; and
(5) Target discrimination.

f. Supervising the firing process by verifying the command solution;

g. Directing the employment of multiple weapons to include:
   (1) Multiple engagement, and
   (2) Salvo tactics.

h. Directing multiple weapons during the pre-enable phase as required;

i. Directing multiple weapons during post-enable phase as required; and

j. Directing completion of engagement IAW CANSAD.

8. **Training Limitations:** Nil

EO 009.01 – DIRECT MULTIPLE SUBMARINE LAUNCHED WEAPONS
009.01.01 – DIRECT MULTIPLE SUBMARINE LAUNCHED WEAPONS PRACTICAL

Location Conditions		**Instructional Sequence**
Fred Sherwood Trainer or Submarine alongside		After PO 008
Time	**Classification**	**Instructor**
1200 min	Secret	Min Qual: AILU Recommended Qual: NWO-SUB with at sea XO experience or submarine Commanding Officer

Learning Objectives	**Method(s)**
1) Respond to Misfire/Hangfire 2) Post Launch Commentary 3) Salvo tactics 4) Controlling Weapons a. Pre-enable b. Post-enable 5) Complete engagement	• EDI

Scenario	Watched Dived at SD
Activity Steps	***Introduction*** Instructor will describe a scenario in which this performance would be done onboard a submarine and go over how the class will be run (5 min) ***EDI*** *Practice* Students will have multiple practical runs to practice independently directing weapon firings as an AC in a tactical environment against multiple hostile surface or subsurface contacts. Instructor will provide a scenario brief to students prior to commencing run encompassing weapon's system status and run mission. Each student will follow correct drills and procedures IAW CANSADS, SUBWOPS, and CCS 876 drills to include: a. manage the CIO, provide quality control to target solutions; b. maintain safety of the submarine WRT the weapon; c. follow and enforce drills described in CANSADS; and d. provide advice to command on weapons engagement.

	AC will evaluate recommendations from the WCCO based on the development of the scenario and properly direct WG commands IAW CANSADs. (WCCO position will be held by students after successful completion of PO 008) Instructor will provide supervision, support and feedback. Students will watch or support each other's runs for training benefits. Support will be reduced as students progress. ***Review (15 min)*** Instructor will: 1) Review any common difficulties from the runs; 2) Have each student state their one main takeaway from this practice; and 3) Ask students if they have any further questions.

Instructor Resources	**References**
1) Secret Hard Drive with reference material 2) WAPP 3) Secret Laptop 4) SFCS with the ability to inject contacts	1) CANSADs (NCPM 412) 2) SUBWOPs 3) CFCD 106

Notes to Instructor
• Instructor will manage PT, BAT, TAT times during trainer time.

EC 009 – DIRECT MULTIPLE SUBMARINE WEAPONS DURING ATTACK

Scope	**Location Conditions**	**Instructional Sequence**
EO 009	Fred Sherwood Trainer or Submarine alongside	After LG 009.01.01
Time	**Classification**	**Instructor**
400 min	Secret	Min Qual: AILU Recommended Qual: NWO-SUB with at sea XO experience or submarine Commanding Officer

Scenario	From run plans.
Assessment Steps	**Input** – Given operational fire control system with ability to inject simulated contacts and conduct synthetic weapons firings. WAPP, CANSADs, CFCD 106, and student threat pack. Instructor will provide a scenario brief to students prior to commencing run encompassing weapon's system status and run mission. **Performance** – Each student will independently direct weapon firings as an AC in a tactical environment against multiple hostile surface or subsurface contacts. Trainee will have 90 min to complete the run. Each student will follow correct drills and procedures IAW CANSADS, SUBWOPS, and CCS 876 drills to include: d. manage the CIO, provide quality control to target solutions; c. maintain safety of the submarine WRT the weapon; e. follow and enforce drills described in CANSADS; and f. provide advice to command on weapons engagement. AC will evaluate recommendations from the WCCO based on the development of the scenario and properly direct WG commands IAW CANSADs. (WCCO position will be held by students after successful completion of PO 008) Instructor will assess based on EC 009 checklist. **Output** – Trainee must achieve 14/17 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment. **Feedback** –Verbal debrief of PC checklist. (10 min per student after run)
Resources	**References**

1) WAPP, 2) CANSADs, 3) CFCD 106, and 4) student threat pack 5) Submarine Fire Control System	1) CANSADS, **2)** SUBWOPS, and **3)** CFCD 106
Notes to Instructor	
Student given the EC checklist to review prior to EC.	
Limitations	Nil
Impact of Failure	IAW NPTORD 4500-54

EC 009 DIRECT MULTIPLE SUBMARINE WEAPONS DURING ATTACK			
Item	**SAT**	**UNSAT**	**Remarks**
Supervise closing-up checks			
Direct warm up weapon	☐	☐	
Direct WCCO to bring weapon system to action state	☐	☐	
Ordering tactics			
Ordering selected tactic	☐	☐	
Order pre-sets applied	☐	☐	
Order select tube	☐	☐	
Order weapon exercised	☐	☐	
Ensure weapon safety criteria is met	☐	☐	**critical**
Verify tactics IAW CFCD 106	☐	☐	
Report to CO ready to fire IAW CANSADs	☐	☐	
Supervise Firing Weapons			
Verify command solution	☐	☐	
Direct firing of multiple weapons			
Salvo firing	☐	☐	**critical**
Target discrimination	☐	☐	**critical**
Direct the employment of weapons post launch			
Identify changes to enabling criteria	☐	☐	
Direct wire guidance commands to include:			
Manipulate fence	☐	☐	
Change post launch pre-sets	☐	☐	
React to countermeasures and false homing	☐	☐	
Completing engagement			
Direct completion of engagement IAW CANSAD.	☐	☐	

Score: /17

Trainee must achieve 14/17 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment.

Positive Feedback	Improvement Feedback

NTDC(A) Approval – 09/02/2022
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Version No: 1.0
MITE Crse ID : 122901

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor **I have discussed the results of this assessment with the trainee.**		**Trainee** **I have been debriefed on the result of this assessment.**	
Assessor	**Name:**	**Date:**	**Signature:**
Trainee	**Name:**	**Date:**	**Signature:**

PROTECTED B (when complete)

PC 009 – DIRECT MULTIPLE SUBMARINE WEAPONS DURING ATTACK

Scope	Location Conditions	Instructional Sequence
PO 009	Fred Sherwood Trainer or Submarine alongside	After EO EC 009.01
Time	**Classification**	**Instructor**
400 min	Secret	Min Qual: AILU Recommended Qual: NWO-SUB with at sea XO experience or submarine Commanding Officer

Scenario	From run plans.
Assessment Steps	**Input** – Given operational fire control system with ability to inject simulated contacts and conduct synthetic weapons firings. WAPP, CANSADs, CFCD 106, and student threat pack. Instructor will provide a scenario brief to students prior to commencing run encompassing weapon's system status and run mission. **Performance** – Each student will independently direct weapon firings as an AC in a tactical environment against multiple hostile surface or subsurface contacts. Trainee will have 90 min to complete the run. Each student will follow correct drills and procedures IAW CANSADS, SUBWOPS, and CCS 876 drills to include: e. manage the CIO, provide quality control to target solutions; d. maintain safety of the submarine WRT the weapon; g. follow and enforce drills described in CANSADS; and h. provide advice to command on weapons engagement. AC will evaluate recommendations from the WCCO based on the development of the scenario and properly direct WG commands IAW CANSADs. (WCCO position will be held by students after successful completion of PO 008) Instructor will assess based on PC 009 checklist. **Output** – Trainee must achieve 14/17 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment. **Feedback** –Verbal debrief of PC checklist. (10 min per student after run)
Resources	**References**

6) WAPP, 7) CANSADs, 8) CFCD 106, and 9) student threat pack 10) Submarine Fire Control System	4) CANSADS, 5) SUBWOPS, and 6) CFCD 106
Notes to Instructor	
Student given PC checklist to review prior to PC.	
Limitations	Nil
Impact of Failure	IAW NPTORD 4500-54

PC 009 DIRECT MULTIPLE SUBMARINE WEAPONS DURING ATTACK			
Item	**SAT**	**UNSAT**	**Remarks**
Supervise closing-up checks			
Direct warm up weapon	☐	☐	
Direct WCCO to bring weapon system to action state	☐	☐	
Ordering tactics			
Ordering selected tactic	☐	☐	
Order pre-sets applied	☐	☐	
Order select tube	☐	☐	
Order weapon exercised	☐	☐	
Ensure weapon safety criteria is met	☐	☐	**critical**
Verify tactics IAW CFCD 106	☐	☐	
Report to CO ready to fire IAW CANSADs	☐	☐	
Supervise Firing Weapons			
Verify command solution	☐	☐	
Direct firing of multiple weapons			
Salvo firing	☐	☐	**critical**
Target discrimination	☐	☐	**critical**
Direct the employment of weapons post launch			
Identify changes to enabling criteria	☐	☐	
Direct wire guidance commands to include:			
Manipulate fence	☐	☐	
Change post launch pre-sets	☐	☐	
React to countermeasures and false homing	☐	☐	
Completing engagement			
Direct completion of engagement IAW CANSAD.	☐	☐	

Score: /17

Trainee must achieve 14/17 SATs INCLUDING ALL CRITICAL FACTORS to pass this assessment.

Positive Feedback	Improvement Feedback

Overall Assessment – Circle One			
Met Standard		**Did Not Meet Standard**	
Assessor **I have discussed the results of this assessment with the trainee.**		**Trainee** **I have been debriefed on the result of this assessment.**	
Assessor	**Name:**	**Date:**	**Signature:**
Trainee	**Name:**	**Date:**	**Signature:**

PROTECTED B (when completed)

CHAPTER 3: ASSESSMENT OF TRAINEE

NTDC(A) Approval – 09/02/2022
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GENERAL

9. All PCs are based on POs in Chapter 2 of this QSP. Trainees must achieve the standard to be awarded the qualification. PCs will be administered in the form of practical or written assessments and the trainee must pass all POs to successfully complete this course.

ENABLING AND PERFORMANCE CHECKS

All ECs are formative assessments that help the trainee and the instructors recognize progress or gaps in learning so that confirmation or corrective action (such as additional practice or remedial instruction) can be provided. ECs also reinforce learning and increase the retention of material for longer periods in preparation for the PC. Non-critical EC results do not factor into a trainee's successful completion of the PO.

The process for critical ECs and PCs is outlined in NPTORD 4500-54. For administrative purposes, a Critical EC has the same importance as a PC.

PCs are summative assessments used to determine if the trainee has met the standard under the conditions (where possible) of the PO. Supplemental PCs are administered upon failure of a PC and must be in the same format and have the same standard as the original PC.

RESPONSIBILITY FOR TESTING

Assessment of trainees is to be completed IAW the standards delineated in this QSP, NPTORD 4500-54, and CFITES Vol 7 – Evaluation.

PROGRESS MONITORING

Monitoring learning is the responsibility of all instructional staff and supervisors. Monitoring, including unsatisfactory progress, shall adhere to CFITES Vol 6 – Conduct of Instructional Programmes.

Continuous monitoring of trainee progress is required to provide the following:

a. Early warning of trainee difficulties;

b. Feedback on the effectiveness of training;

c. Information for the Training Review Board (TRB) (if required); and

d. Information to assist in the completion of the CF 377 Course Report.

Trainees experiencing difficulty in any area of performance are to be counselled with regard to the nature of their shortcomings and suggested corrective action IAW TE SOs/SOPs and DAOD 5516-5.

PC failures and supplemental PC failures are to be handled IAW TE SOs/SOPs and NPTORD 4500-54. Supplemental ECs and PCs shall not be administered within 24 hours of the trainee being informed of their EC or PC failure. Trainees will have 24 hours at minimum to prepare for a retest.

TRAINING REVIEW BOARD

The aim of the TRB process is to review a trainee's suitability for continued training and education, while ensuring that the trainee is afforded a fair and just assessment of all relevant information before a decision is made. TRBs shall be convened and conducted IAW NPTORD 4500-54.

COURSE REPORTS

Trainees shall be graded pass or fail IAW NPTORD 4500-51.

Course reports (CF 377) shall be prepared and distributed IAW NPTORD 4500-51. If applicable, training limitations and/or incomplete training must be included in the course report.

Narratives shall indicate trainee's strengths and weaknesses based on PC results to provide information to the Career Manager.

A record of trainee counselling shall be maintained for 2 years by the TE IAW Information Management Standard (IMS) 6001-2-1 and Defence Subject Classification and Disposition System (DSCDS) Primary Number 4500.

Task List

1. Tasks retrieved from A-PD-055-002/PP-001 Naval Warfare officer Occupational Specification dated 2018-07-18 and Submarine Operations Room Officer OSS A-PD-055-003/PQ-001 dated 2010-08-11.

Task #	Task Statement	Trg / No Trg	PO
BT 0010	Schedule the ship's program	Trg	001
BT 0030	Plan an anti-surface operation	Trg	001
BT 0050	Plan an UWW operation	Trg	001
BT 0100	Manage anti-surface warfare operation	Trg	003/004
BT 0110	Manage UWW operation	Trg	003/004
BT 0130	Prepare operational messages	Trg	002
BT 0140	Prepare evaluation operational reports	Trg	002
BT 0160	Supervise anti-surface weapons firing analysis	No Trg	
BT 0170	Supervise UWW weapons firing analysis	No Trg	
BT 0300	Implement anti-surface warfare tactics	Trg	003/004
BT 0310	Implement UWW tactics	Trg	003/004
BT 0330	Advise command on anti-surface warfare safety	Trg	004
BT 0340	Advise command on UWW safety	Trg	004
BT 0500	Manage intelligence	Trg	004
CT 0120	Advise command on operating and fighting efficiency of submarine	Trg	001
T 0001	Surface the submarine	Trg	003
T 0002	Dive the submarine	Trg	003
T 0003	Plan submarine program	Trg	001
T 0004	Direct submarine routines surfaced and dived	Trg	004
T 0005	Direct Emergency Operating Procedures (EOPs)	Trg	004
T 0006	Direct Damage Control Organization	Trg	006
T 0007	Manage the Manning/Maintenance/Operationg efficiency of the submarine	No Trg	
T 0008	Direct Blind Pilotage	No Trg	
T 0009	Direct tactical maneuvering	Trg	004
T 0010	Direct combat system Line-Ups	Trg	004

T 0011	Direct submarine attack team activities	Trg	008
T 0012	Direct employment of submarine weapons	Trg	008
T 0013	Direct intelligence, surveillance, and reconnaissance operations (ISR)	Trg	002
T 0014	Execute Submarine Operations	Trg	004
T 0015	Coordinate submarine logistics	No Trg	
T 0016	Coordinate staff work	No Trg	
T 0017	Direct Combat Information Organization	Trg	003/004
T 0018	Direct Force protection	Trg	005
T 0019	Asses the threat	Trg	002
T 0020	Apply Rules of Engagement (ROE)	Trg	
T 0021	Make recommendations to command	Trg	
T 0022	Interpret meteorological, oceanographic and acoustic environmental information	Trg	
T 0023	Identify incorrect/abnormal conditions	Trg	
T 0024	Interpret the Target Motion Analysis (TMA) plots	Trg	
T 0025	Assign contact tracking priorities	Trg	
T 0026	Execute periscope safety techniques/ Eyes-only Attack	Trg	
T 0027	Administer production of records	Trg	
T 0028	Effect Submarine Combat Readiness	Trg	
T 0029	Plan Submarine Operations	Trg	

Record of Proceedings

Dates: 10 – 21 SEP 2019

Location: CFB Halifax, Building S-17

Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Novitsky	R.G.	LCdr	00207	NTDC (A)
Matheson	W.A.	LCdr	00207	HMCS Windsor
Mercer	P.F.	Lt(N)	00207	HMCS Windsor
Nguyen-Huynh	H.T.	Lt(N)	00207	CANSUBFOR
Guthrie	N.A.	SLt	00211	NTDC (A)
Schertzer	J.	SLt	00207	HMCS Windsor

Working Group Objective:

1. The objective of this working group is to rebuild the SMORO course.

Specification Task – No Train Decisions:

Task #	Task Statement	Substantiation
T 0007	Manage the manning/maintenance/operating efficiency of the submarine	In PO 001
T 0008	Direct blind pilotage	This is mostly a quality assurance task that the board felt was experience based.
T 0015	Coordinate submarine logistics	This is an XO task, not required day one on the job. Board believed training would fade before need to complete task would be required. Can be learned on-the-job.
T 0016	Coordinate staff work	This is an XO task, not required day one on the job. Board believed training would fade before need to complete task would be required. Can be learned on-the-job.

Discussions / Decisions:

2. Members discussed adding task. Task discussed were Operation Honour, NATO publications, SRF, recce, and center of gravity.

3. The importance of Operation Honour was discussed and how to word it as a task. Board had could not fit it into an existing PO and the army ethics PO (PO 99) was discussed. Board investigated if there was higher navy direction that mandated Operation Honour to be included in the course and

was told there was no direction. Board decided that although important material was covered in other places and it was not included in the course.

4. Knowledge encompassing NATO publications, SRF, recce, and center of gravity would fall under other task or previous training.

5. Board worked on grouping tasks into Performance Objectives (POs). Many tasks were grouped under a tentative PO called “Run a Dive Watch”. TDO asked about the viability of multiple POs, each covering an individual submarine tasking or operation verses one PO covering it all. A board member warned against compartmentalizing taskings/POs as the SMORO job is multifaceted. Other members agreed. It was decided that “Run a Dive Watch” would be divided into two POs. “Run a Dive Watch in Support of Force Generation Operations” (Similar to OTT1) and “Run a Dive Watch in Support of Force Employment Operations” (Similar to OTT2). This would allow some focus on different aspects of the job without compartmentalizing having a negative effect on the realism of the POs. First PO would focus on safety while second on warfare.

6. Board debated on how to incorporate EOPs. It was decided that this course will not focus on EOPs due to lake of ability to simulate in school trainer (Fred Sherwood) .

7. Board discussed what is required of an SMORO as it pertains to the operational planning process. It was decided that part of the job was creating a duty plan for a sailing period. All steps of preparing the plan were detailed in the PO standard statement as well as the headings required in a duty plan.

8. Question raised about the requirement to instruct “interpreting meteorological oceanographic and acoustic environmental info” (T 0022) task. It was decided to include in course since some members never receive this training.

9. The requirement to have a force protection component to the course was discussed. The SMORO role in force protection is small, but board felt it was important to cover due to the risk involved in a failure in this area. A PO was created “Advise Command on Force Protection Requirements” as to have the students participate in a force protection exercise as a PC would be impractical. The performance objective would be for the student to present a force protection brief.

10. The board debated including the task “Direct the Damage Control (DC) Organization” (T 0006) as it was an XO task and not required day one on the job. Also, the concern around skill fade was discussed and the efficiency of including this training at this time. It was brought up that no other training may be provided for students before being appointed XO. Having another course or onboard package to complete sometime after the SMORO course to qualify for XO was discussed, but the board felt there would not be support for this. It was decided a PO called “Respond to a Submarine Emergency as the Incident Controller” would be created. It would be an exposure to the duties of an IC, as running students through scenarios in a trainer would be impractical and not an efficient use of resources. Although not ideal the board felt that due to the safety concern the exposure was necessary.

11. Seamanship evolutions brought up similar issues as acting as Incident controller. This is an XO tasks not required day one on the job. However, the board was concerned that this would be the only exposure a student may have before being required to complete the task. It was brought up that a safety infraction was seen onboard and it was felt that it was due to lack of experience. A PO “Conduct Submarine Seamanship Evolutions” was created due to the safety concerns. As

seamanship evolutions will not be able to be done at the school, they will be done as table top exercises with safety considerations included.

12. The TDO's concern with training related to the XO position is that it is not required day one on the job. Skill and knowledge could fade by the time a graduate was in the XO position. TDO will investigate the possibility of reviewing the XO's responsibilities and the need for a XO course that is just-in-time training.

13. A PO Employ submarine weapons was created due to possible modulation of this task. Meets command intent to reinvigorate weapons culture in the fleet.

14. It was decided by the board that diving and surfacing the submarine would not be tested on a PC. It will be covered as an EO and tested as an EC in PO 003. Adding this to the PC would not fit practically into a realistic scenario. Board believes that the tasks is simple enough that a hard (summative) assessment is not required.

15. Maximum and minimum course loading was proposed to be lower than the current QSP. This is due to the historical numbers, there is not usually more than 4 individuals available for this training. The board members felt that the lower numbers will support better training and the TDO agreed referencing CFITES Vol 5, Annex D, and Instructor to Learner Ratios. Although students could be working on other projects or watching fellow students practice in the trainer, trainer time would take one instructor to one student.

Guidance to Development and/or Instructional Staff:

16. .

Follow Up Action:

Item	Action	OPI	Expected Date of Completion

Dates: 1-12 March 2021
Location: Halifax, NS
Working Group Members:

Surname	Initials	Rank	MOSID	Unit
Collins	PN	LCdr		NTDC(A)
Guthrie	NA	Lt(N)		NTDC(A)
Holland	CD	Cdr		CANSUBFOR
Hendry	DI	LCdr		HMCS Chicoutimi
Han	H	Lt(N)		NFSA

Working Group Objective:

1. The purpose of the working group is to update Submarine Warfare Director (SMWD), SMORO, and CRWK Dived QSP to reflect the new submarine torpedo. The working group was also tasked to find a solution to gaps in weapons training for submariners who do not receive the Warfare Director course as well as develop refresher training.

2. Board chair introduced local members and advised that west coast members will be joining by Microsoft Teams at 1300 local time. TDO gave a systems approach to training brief and outlined the agenda and goals of the working group.

3. Working Group convened 1 March 2021 at CFB Halifax.

4. Working group used PO and PC checklist created for SMWD PO (Operate Weapon) to update SMORO PO (Direct Employment of Submarine Weapons). Board discussed focusing on WCCO and not assessing AC as stated in current PO. TDO brought up that current PO indicates AC, but checklist appears to focus on WCCO. However, removing AC moves away from the intent of the SMORO QSP Board. After discussion it was decided to create two POs, one for the WCCO (Operate Submarine Weapon) and the other for AC (Direct Submarine Weapon). Both positions would be practiced at the same time in developmental runs, with students acting as both AC and WCCO. This will reduce course and trainer time. PCs will be done separate. During the WCCO PC the instructor will act as AC and during the AC PC students successful in the WCCO PC will act as WCCO.

5. WG working on revising SMORO POs. It was decided that the SMORO would need to complete more complex runs using multiple weapons verse the SMWD how would operate a single weapon. POs and PCs were written for both AC and WCCO to reflect the requirement to operate or direct multiple weapons.

Course History:

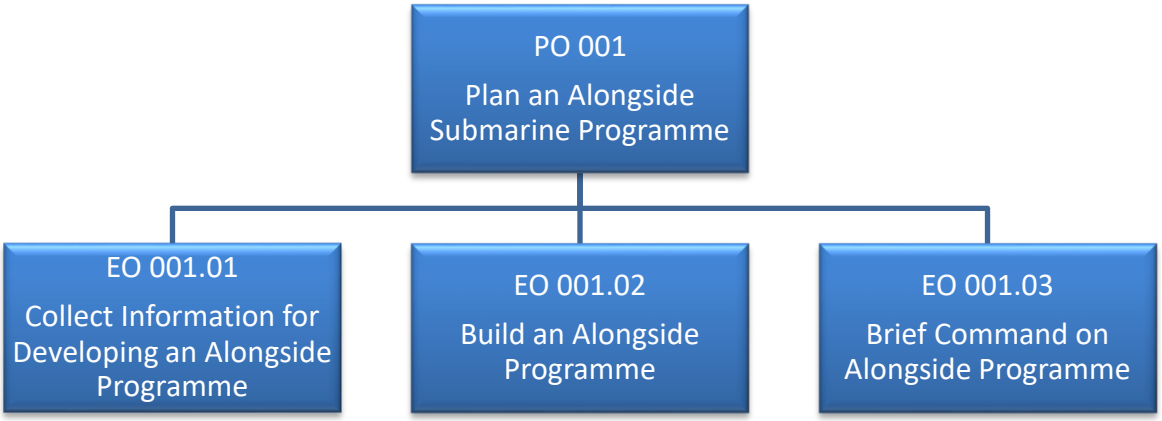
The SMORO (AILU) course was the offspring of a course that was taught in Canada as Submarine Weapons and Operations course and was last run in 1998 with three students. This was previously used as the primary means to train Combat Officers (CBTO) for the submarines. Perspective XOs had to attend an Australian XO (or Attack Coordinators) course. In 2000, it was decided to stop running the SWO course in Canada until all of the newly purchased VICTORIA class submarines were brought to steady state in Canada. However, as time passed the SMWOC (Submarine Warfare Officers Course) in Australia was used to qualify our perspective Combat Officers and the ACs course was continued to be used to qualify our perspective XOs. In 2003 a QSP board was sat for the SMORO (Submarine Operations Room Officer) which would, in concept, qualify perspective submarine CBTO and XOs and be tied to the surface ORO course. The course was first run in 2008 with two students and then again in 2009 with three students as a yearlong course. Given the small number of students, no pilot course was done and that there are no instructor's notes, there is no recorded course history. In late 2009, a QSP board was convened

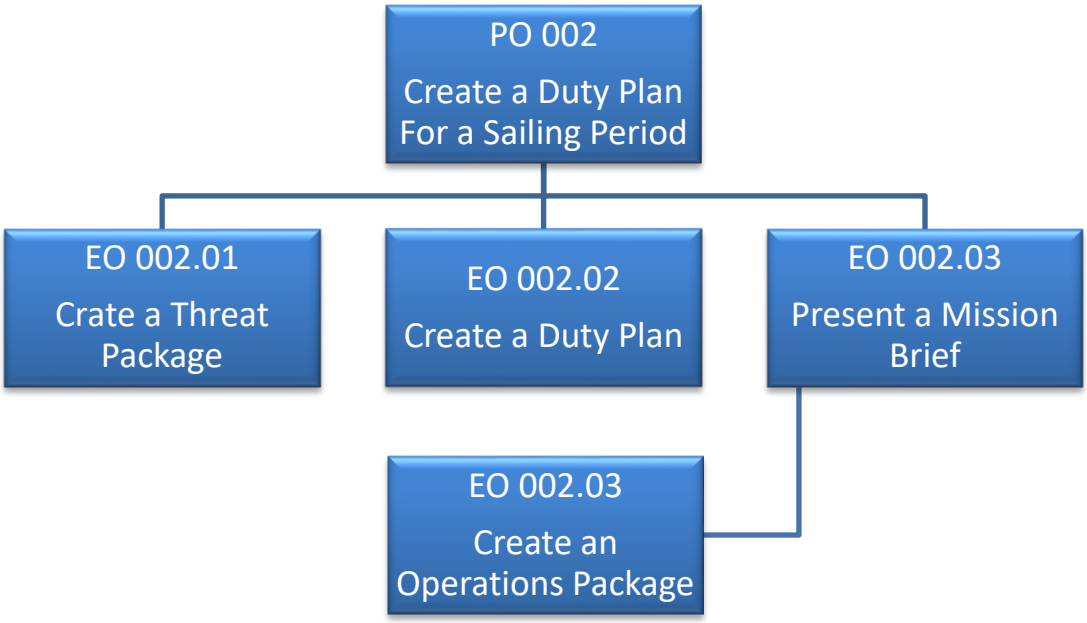
for the Surface ORO course, which by extension triggered a QSPWB in March 2010 for the SMORO. The board essentially had a blank sheet from which to start constructing the course to qualify perspective submarine CBTOs and XOs.

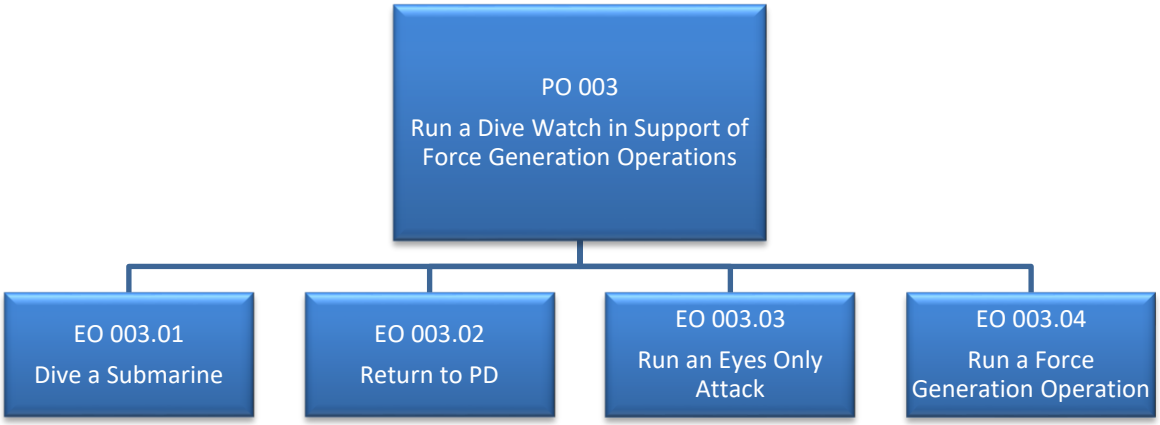
Aug 2014 – In response to Commander MARPAC's Initiating Directive to align training with requirements, a QSP Review Board (QSPRB) was convened at CFNOS to eliminate the Surface ORO training not applicable to Submariners. Training was reviewed IAW the QSPRB Record of Proceedings, and the course was reduced from 220 training days to approximately 93 training days.

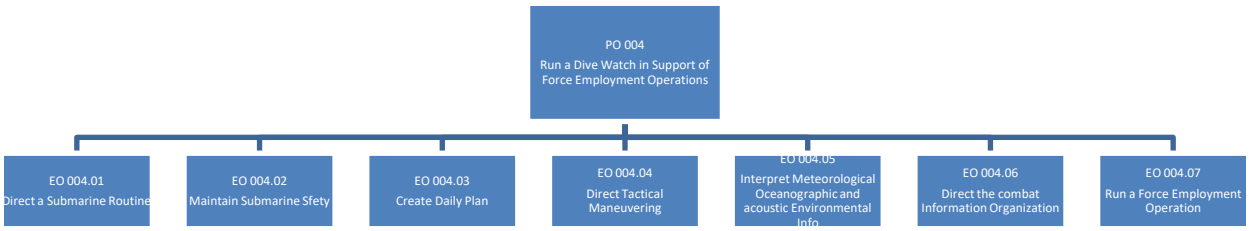
Mar 2015 – The QSP was reviewed by the Commander Canadian Submarine Force (CCSF) and minor issues were identified as needing amending. CCSF approved the amendments and on 25 Mar 2015 and amendments were made. Amendments include adding Submarine Sonar Officer (AILT) to the list of prerequisites in Ch. 1; adding Maritime Component Commander (MCC), SUBOPAUTH, and Commander Canadian Submarine Force (CCSF) under national command structure in CH 4; and other minor amendments. The authority for CH 1&2 was changed to CNPTG and the authority for CH 3&4 was changed t

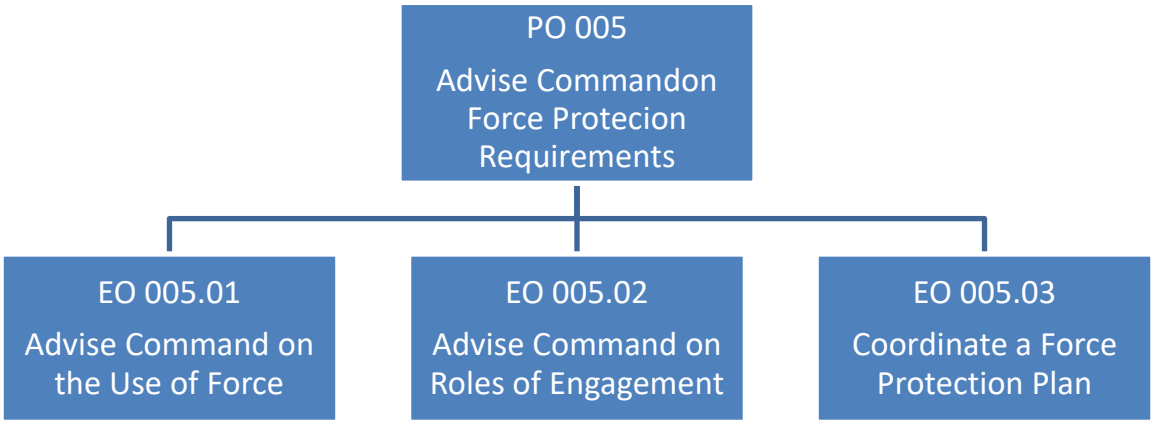
QSP Scalars

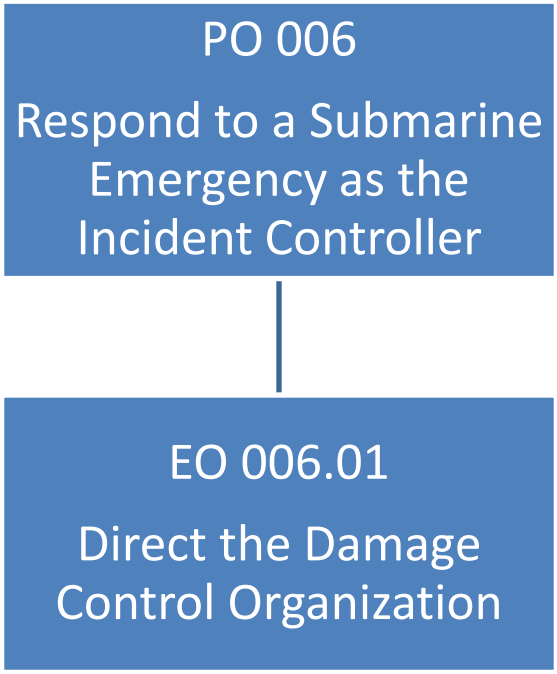

PO 001
Plan an Alongside Submarine Programme
EO 001.01
Collect Information for Developing an Alongside Programme
EO 001.02
Build an Alongside Programme
EO 001.03
Brief Command on Alongside Programme

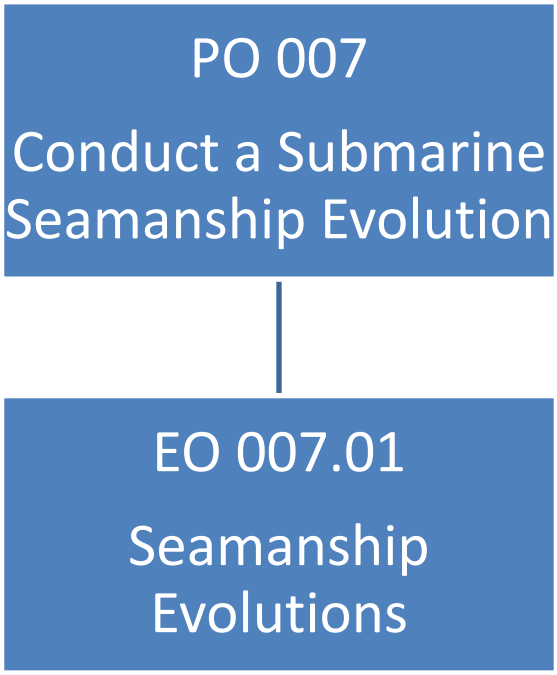

PO 002
Create a Duty Plan For a Sailing Period
EO 002.01
Crate a Threat Package
EO 002.02
Create a Duty Plan
EO 002.03
Present a Mission Brief
EO 002.03
Create an Operations Package

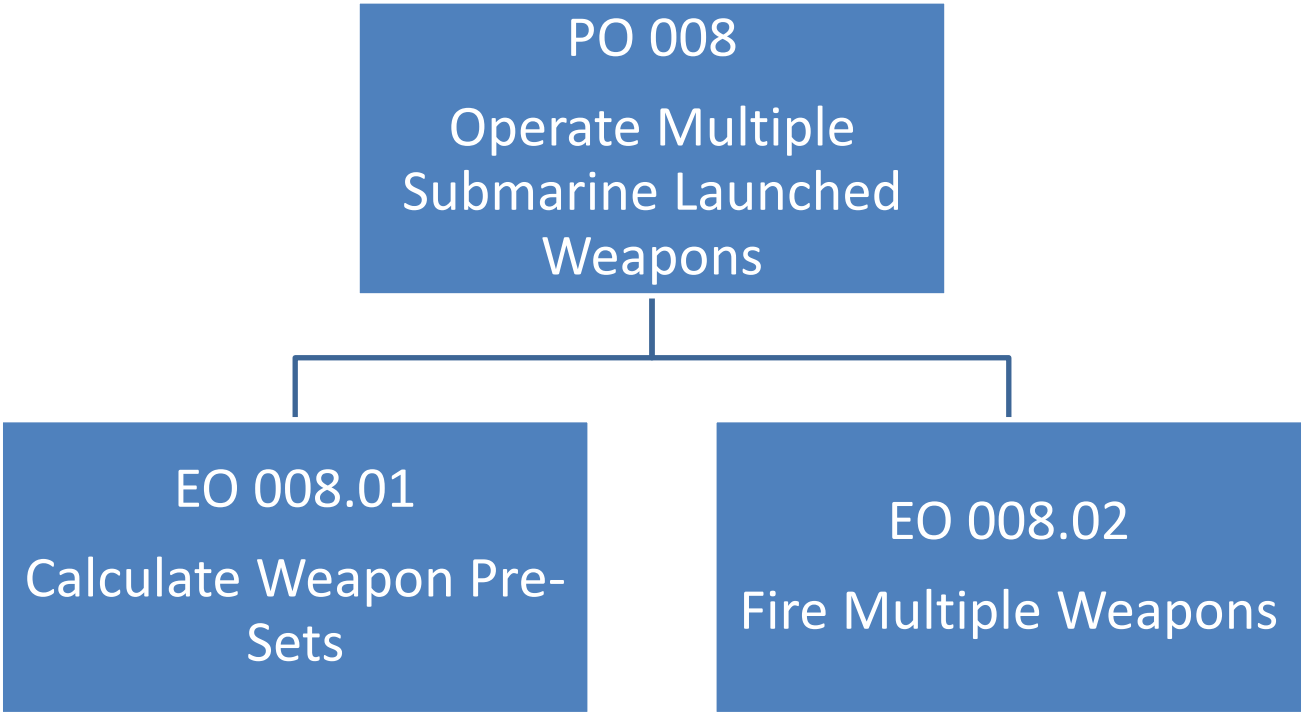

PO 003
Run a Dive Watch in Support of Force Generation Operations
EO 003.01
Dive a Submarine
EO 003.02
Return to PD
EO 003.03
Run an Eyes Only Attack
EO 003.04
Run a Force Generation Operation

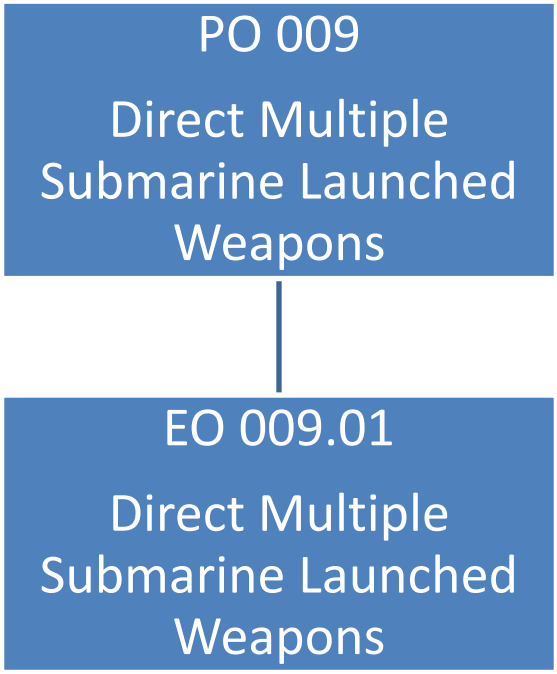

PO 004
Run a Dive Watch in Support of Force Employment Operations
EO 004.01
Direct a Submarine Routine
EO 004.02
Maintain Submarine Sfety
EO 004.03
Create Daily Plan
EO 004.04
Direct Tactical Maneuvering
EO 004.05
Interpret Meteorological Oceanographic and acoustic Environmental Info
EO 004.06
Direct the combat Information Organization
EO 004.07
Run a Force Employment Operation


PO 005
Advise Commandon Force Protecion Requirements
EO 005.01
Advise Command on the Use of Force
EO 005.02
Advise Command on Roles of Engagement
EO 005.03
Coordinate a Force Protection Plan


PO 006
Respond to a Submarine Emergency as the Incident Controller
EO 006.01
Direct the Damage Control Organization


PO 007
Conduct a Submarine Seamanship Evolution
EO 007.01
Seamanship Evolutions


PO 008
Operate Multiple Submarine Launched Weapons
EO 008.01
Calculate Weapon Pre-Sets
EO 008.02
Fire Multiple Weapons


PO 009
Direct Multiple Submarine Launched Weapons
EO 009.01
Direct Multiple Submarine Launched Weapons

Costing Information (Estimated)

Include information about cost in whichever format is applicable (**Not to be deleted**).

Costing Annex_Template.xlsx

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: SMORO		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Airfare (Return)	4	$1,000.0	N/A	$4,000	$4000	$
R&Q	4	$78.75	66	$20,790	$20,790	$
Incidentals	4	$17.3	66	$4,567.20	$4,567.20	$
Subtotal				**$29,357**	**$29,357**	**$**
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
PP&S (Estimate)	4	$50.00	1	$200	$200	$
Classroom (square meters)				…	…	…
Computer requirements		$	1	$	$0	$
IT Equip (Proj, Smart board etc)		$	1	$	$0	$
DWAN connection (Set up)		$	1	$	$0	$
Link Toolkit (on DVD)		$	1	$	$	$
Instructional Materials (Printed / Handouts etc)		$	1	$	$	$
Subtotal				**$**	**$**	**$**
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
Instructor (PO1/WO) (incremental)2		$		$	$	$
Incremental Civilian Instructor						
Subtotal				**$**	**$**	**$**
	Total Estimated Costs			**$**	**$**	**$**

Assessment Plan

EC/PC	Scope Covered	Purpose	How	Pass/Fail Criteria
EC 001.02A	EO 001.02A	To assess student's competence in building an alongside programme.	Practical	Student must obtain 5 / 5 SATs IAW PC 001.01 Section 1, Checklist
EC 001.02B	EO 001.02B	To assess student's competence briefing command on an alongside programme.	Practical	Student must obtain 3 / 3 SATs IAW PC 001.01 Section 2, Checklist
PC 001	PO 001	To assess student's competence in planning an alongside submarine programme.	Practical	Student must obtain 11 / 11 SATs IAW PC 001 Checklist
EC 002.01	EO 002.01A - C	To assess student's competence presenting a mission brief.	Practical	Student must obtain 5 / 5 SATs IAW PC 002 Section 1, Checklist
PC 002	PO 002	To assess student's competence in creating a duty plan for a sailing period.	Practical	Student must obtain 45 / 45 SATs IAW PC 002 Checklist
EC 003.02	EO 003.02	To assess student's competence in returning to periscope depth.	Practical	Student must obtain 26 / 26 SATs IAW EC 003.02 Checklist
EC 003.03	EO 003.03	To assess student's competence in conducting an eye's only attack.	Practical	Student must obtain 6 / 7 SATs IAW EC 003.03 Checklist
EC 003.04	EO 003.04	To assess student's competence running a force gerneration operation.	Practical	Student must create a Briefing Note covering all LOs in LG 003.04A and fill fill out records examples listed in LG 003.04B LOs including completion of the GOP will IAW References
PC 003	PO 003	To assess student's competence running a dive watch in support of force generation operations.	Practical	Student must obtain 24 / 27 SATs IAW PC 003 Checklist
EC 004.02	EO 004.02	To assess student's competence in retruning to periscope depth.	Practical	Students must return to periscope depth IAW CANSADS and NCPM 401(B)
EC 004.03	EO 004.03	To assess student's competence in creating a daily plan.	Practical	Students must complete a Daily Plan that incudes the events of the scenario given.
EC 004.04	EO 004.04	To assess student's competence in directing tactical maneuvering.	Practical	Student must create a briefing note explaining the LOs in LG 004.04.

EC 004.06	EO 004.06	To assess student's competence in directing the combat information organization.	Practical	Students must create a Briefing Note covering the items listed in EO 004.06 Activity Steps
EC 004.07	EO 004.07	To assess student's competence in running a force employment operation.	Practical	Student must create a Briefing Note outlining EO 004.07 LOs.
PC 004	PO 004	To assess student's competence in Runing a dive watch in support of force employment operations.	Practical	Student must obtain 33 / 36 SATs and all critical factors IAW PC 004 Checklist
EC 005.03	EO 005.03	To assess student's competence in coordinating a force protection plan.	Practical	Student must create a FP Intentions message IAW References.
PC 005	PO 005	To assess student's competence advising command on force protection requirements.	Practical	Student must obtain 7 / 7 SATs IAW PC 005 Checklist
EC 006.01	EO 006.01	To assess student's competence in directing the damage control organization.	Practical	Student must obtain 10 / 12 SATs IAW EC/PC 006 Checklist
PC 006	PO 006	To assess student's competence responding to a submarine emergency as the incident controller.	Practical	Student must obtain 10 / 12 SATs IAW EC/PC 006 Checklist
EC 007	EO 007	To assess student's competence in seamenship evolutions.	Practical	Students must create a Briefing Note covering all points in the EC/PC 007 Checklist.
PC 007	PO 007	To assess student's competence in conducting a submarine seameanship evolution.	Practical	Student must obtain 5 / 6 SATs IAW EC/PC 007 Checklist
EC 008	PO 008	To assess student's competence in operating multiple submarine launched weapons.	Practical	Student must obtain 25 / 28 SATs including all criticals IAW EC 008 Checklist
PC 008	PO 008	To assess student's competence in operating multiple submarine launched weapons.	Practical	Student must obtain 25 / 28 SATs including all criticals IAW PO 008 Checklist
EC 009	PO 009	To assess student's competence in directing multiple submarine launched weapons.	Practical	Student must obtain 14 / 17 SATs including all criticals IAW EC 009 Checklist
PC 009	PO 009	To assess student's competence in directing multiple submarine launched weapons.	Practical	Student must obtain 14 / 17 SATs including all criticals IAW PO 009 Checklist

TEL Development Opportunities

Title	Type	Description of Requirement	Lesson	SOR #
Printed Submarine		3D printed sub to table top seamanship and DC evolutions	POs 6-7	

ANNEX G – SCHEDULING INFORMATION

This table summarizes the conditions cell in each LG. The order within the table reflects the order of instruction. All timings are based on maximum load. This is required for the Scheduling Cell and completion is mandatory.

Lesson		Classroom				IT Assets (Trainees & Instructors)					
#	Time (minutes)	Location	Classification	White Board	Desks	DWAN	CSNI	Secret Stand Alone	GPNet	Projector or TV	TEL & Other
PO 001 – PLAN AN ALONGSIDE SUBMARINE PROGRAM											
LG 001.01A	50	Classroom	Confidential	1	4	5	-	-	-	1	Printer
LG 001.01B	50	Classroom	Unclass	1	4	5	-	-	-	1	Printer
LG 001.01C	800	CFMWC	Unclass	-	4	-	-	-	-	1	-
LG 001.02A	100	Classroom	Unclass	1	4	5	-	-	-	1	Printer
LG 001.02B	50	Classroom	Unclass	1	4	5	-	-	-	1	Printer
LG 001.03A	100	Classroom	Unclass	1	4	5	-	-	-	1	Printer
PC 001	600	Classroom	Unclass	1	4	5	-	-	-	1	Printer
PO 002 – CREATE A DUTY PLAN FOR A SAILING PERIOD											
LG 002.01A	50	Classroom	Confidential	-	4	2	2	-	-	-	Trinity
LG 002.01B	50	Classroom	Confidential	-	4	2	2	-	-	-	Trinity
LG 002.01C	100	Classroom	Confidential	-	4	2	2	-	-	-	Trinity
LG 002.02A	50	Classroom	Confidential	1	4	-	-	-	-	-	
LG 002.02B	150	Classroom	Confidential	1	4	-	-	-	-	-	
LG 002.02C	100	Classroom	Confidential	-	4	2	2	-	-	-	
LG 002.03A	50	Classroom	Confidential	1	4	2	2	-	-	-	Confidential Laptop
LG 002.03B	400	Classroom	Confidential	1	4	2	2	-	-	-	Trinity
PC 002	1000	Classroom	Confidential	1	4	2	4	-	-	1	Classified Stand Alone Laptop with printer / Trinity resources
PO 003 – RUN A DIVE WATCH IN SUPPORT OF FORCE GENERATION OPERATIONS											
EO 003.01A	100	Classroom	Unclass	1	4	5	-	-	-	1	Printer
EO 003.01B	50	Classroom	Unclass	1	4	5	-	-	-	1	Printer
EO 003.02	2000	Fred Sherwood Trainer	Confidential	-	-	-	-	-	-	-	

Lesson		Classroom				IT Assets (Trainees & Instructors)					
#	Time (minutes)	Location	Classification	White Board	Desks	DWAN	CSNI	Secret Stand Alone	GPNet	Projector or TV	TEL & Other
EO 003.03	4000	Fred Sherwood trainer/ Classroom	Confidential	-	-	-	-	-	-	-	
LG 003.04A	400	Classroom	Confidential	1	4	2	4	-	-	1	Classified Stand Alone Laptop
LG 003.04B	150	Classroom	Unclass	1	4	5	-	-	-	1	Printer
LG 003.04C	100	Classroom	Unclass	1	4	5	-	-	-	1	Printer / ECPINS Laptop
PC 003	800	Fred Sherwood Trainger	Confidential	-	-	-	-	-	-	-	
PO 004 – RUN A DIVE WATCH IN SUPPORT OF FORCE EMPLOYMENT OPERATIONS											
LG 004.01A	50	Classroom	Unclass	1	4	5	-	-	-	1	Printer
LG 004.02A	2000	Fred Sherwood Trainer	Confidential	-	-	-	-	-	-	-	
LG 004.03	100	Classroom	Unclass	1	4	5	-	-	-	-	Printer
LG 004.04	100	Classroom	Confidential	1	4	5	4	-	-	1	Printer / Classified Stand Alone Laptop with printer
LG 004.05	400	Classroom	Secret	1	-	-	-	-	-	1	Laptop and projector for Secret presentations
LG 004.06	100	Classroom	Unclass	1	4	5	-	-	-	1	printer
LG 004.07	800	Classroom		1	4	2	4	-	-	1	
PC 004	2400	Fred Sherwood Trainer	Confidential	-	-	-	-	-	-	-	
PO 005 – ADVISE COMMAND ON FORCE PROTECTION REQUIREMENTS											
LG 005.01	200	Classroom	Confidential	1	4	2	4	-	-	1	Classified Stand Alone Laptop /CSNI Trinity Resources

Lesson		Classroom				IT Assets (Trainees & Instructors)					
#	Time (minutes)	Location	Classification	White Board	Desks	DWAN	CSNI	Secret Stand Alone	GPNet	Projector or TV	TEL & Other
LG 005.02	100	Classroom	Confidential	1	4	2	4	-	-	1	Classified Stand Alone Laptop /CSNI Trinity Resources
LG 005.03	100	Classroom	Confidential	1	4	2	4	-	-	1	Classified Stand Alone Laptop /CSNI Trinity Resources
PC 005	200	Classroom	Confidential	1	4	2	4	-	-	1	Classified Stand Alone Laptop /CSNI Trinity Resources
PO 006– RESPOND TO A SUBMARINE EMERGENCY AS THE INCIDENT CONTROLLER											
LG 006.01	400	MCT/SCT Trainer or Classroom	Unclass	-	-	-	-	-	-	-	
PC 006	200	MCT/SCT Trainer or Classroom	Unclass	-	-	-	-	-	-	-	
PO 007– CONDUCT SUBMARINE SEAMANSHIP EVOLUTIONS											
LG 007.01A	200	Classroom	Unclass	1	4	5	-	-	-	1	Printer
PC 007	100	Classroom	Unclass	1	4	5	-	-	-	1	Printer
PO 008 – OPERATE MULTIPLE SUBMARINE LAUNCHED WEAPONS											
LG 008.01.01	300	Classroom	Secret	1	4	-	-	-	-	1	Powerpoint for Secret presentation / Secret Laptop
LG 008.02.01	400	Fred Sherwood Trainer / Or Submarine Alongside	Secret	-	-	-	-	-	-	-	
EC 008	400	Fred Sherwood Trainer / Or	Secret	-	-	-	-	-	-	-	

Lesson		Classroom				IT Assets (Trainees & Instructors)					
#	Time (minutes)	Location	Classification	White Board	Desks	DWAN	CSNI	Secret Stand Alone	GPNet	Projector or TV	TEL & Other
		Submarine Alongside									
PC 008	400	Fred Sherwood Trainer / Or Submarine Alongside	Secret	-	-	-	-	-	-	-	
PO 009 – DIRECT MULTIPLE SUBMARINE LAUNCHED WEAPONS											
LG 009.01.01	1200	Fred Sherwood Trainer or Submarine Alongside	Secret	-	-	-	-	-	-	-	
EC 009	400	Fred Sherwood Trainer or Submarine Alongside	Secret	-	-	-	-	-	-	-	
PC 009	400	Fred Sherwood Trainer or Submarine Alongside	Secret	-	-	-	-	-	-	-	

Plan, Train, Do - NA